

데이터베이스관리

컴퓨터소프트웨어공학과

담당교수: 김희숙
(jasmin11@hanmail.net)

데이터베이스관리

2주차

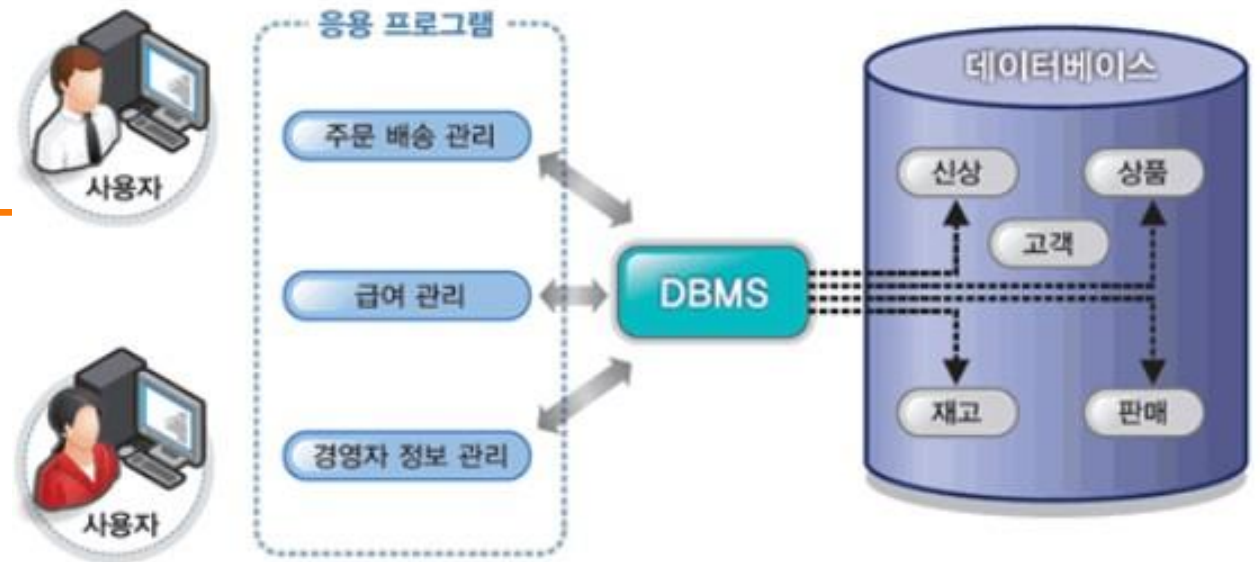
담당교수: 김희숙
(jasmin11@hanmail.net)

데이터베이스



- 데이터베이스:

- 관련된 데이터 모임
- 한 조직의 여러 응용시스템들이 데이터를 공유할 수 있도록 통합, 저장된 운영데이터의 집합체



(그림 출처: "Understanding of Database", 이상구외 공저, 이한, 2012)

테이블(릴레이션)



- 테이블(table): 행과 열로 구성된 2차원의 표 형태

테이블 릴레이션

학생

필드 (속성, 열, 애트리뷰트) column

번호	이름	학년	분반	학과번호
1	하지혜	1	YB	1
2	이정우	1	YA	1
3	오지영	2	J1	2
4	강재미	1	YB	1
5	박철호	2	J1	2

레코드
(튜플, 행)
row

학생(번호, 이름, 학년, 분반, 학과번호)

MySQL 실습

담당교수: 김희숙
(jasmin11@hanmail.net)

[실습] MySQL

- 1단계: 데이터베이스 생성(스키마 생성)
- 사용할 데이터베이스 선택
- 2단계: 테이블 생성
- 3단계: 데이터 입력

[테이블 명세서] ↗

테이블이름: 학과

테이블 이름	열 이름	데이터 형식	제약조건	기본키	외래키	FK테이블	FK열
학과	학과번호	int	NOT NULL	PK			
	학과명	varchar(50)					

학과(학과번호, 학과명)

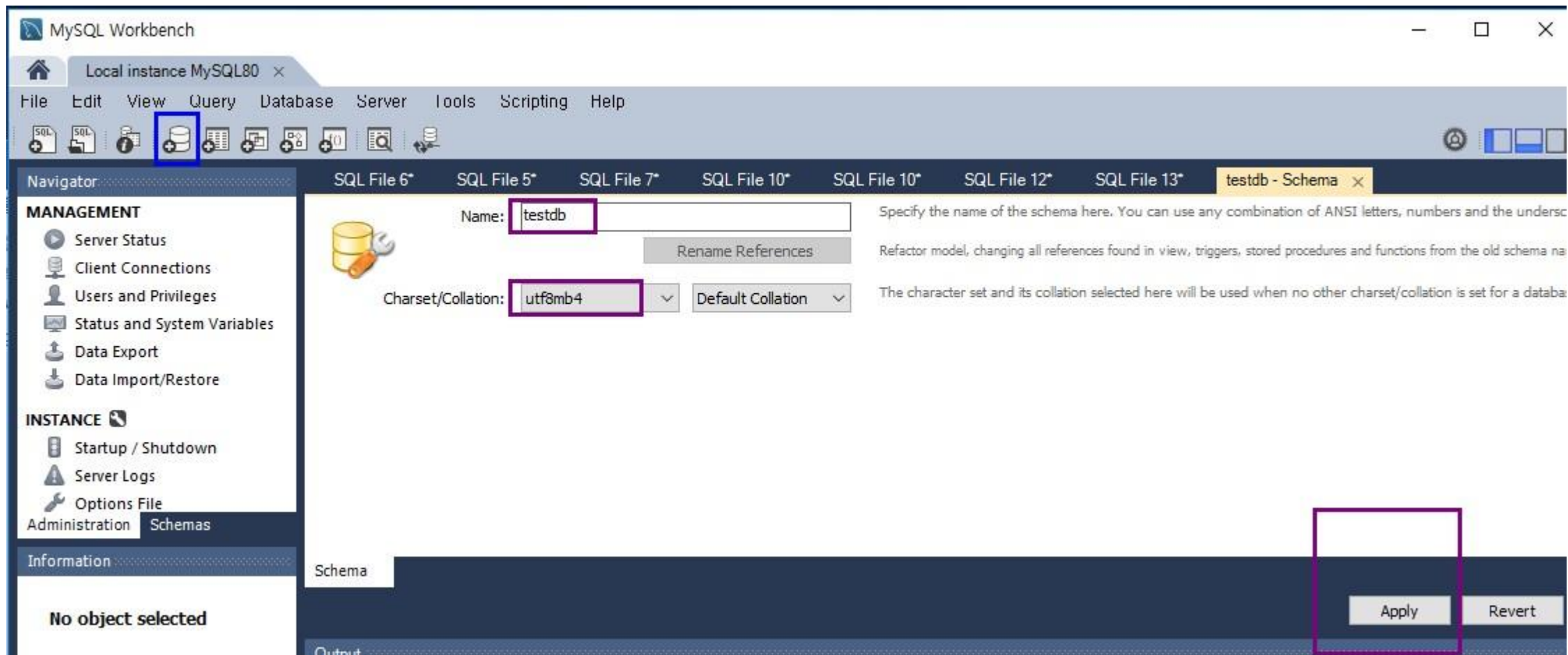
학과 테이블 <<출력 형태>>

학과

학과번호	학과명
1	컴퓨터소프트웨어공학과
2	컴퓨터정보공학과
3	정보통신과

[실습] MySQL

1단계: 스키마 생성 testdb



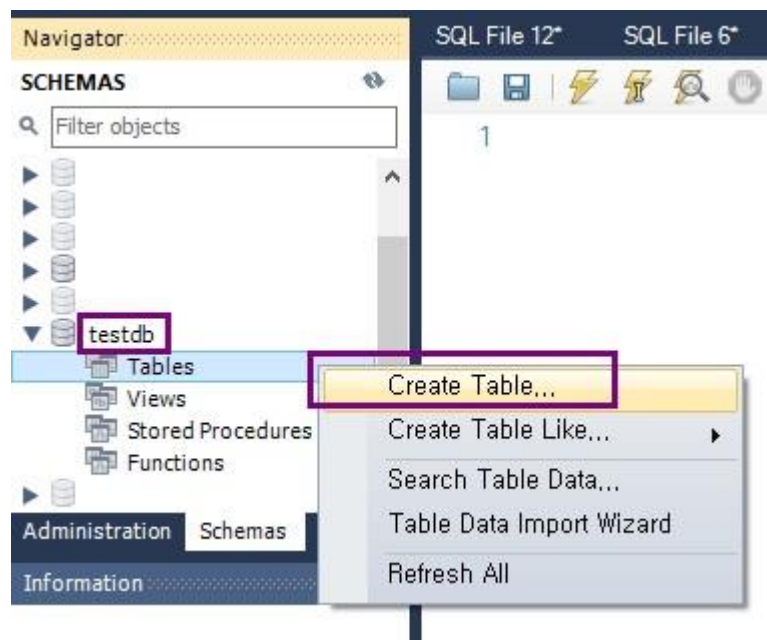
[실습] MySQL

2단계: 스키마 생성

학과

학과

학과번호	학과명
1	컴퓨터소프트웨어공학과
2	컴퓨터정보공학과
3	정보통신과



[실습] MySQL

2단계: 스키마 생성

학과

학과

학과번호	학과명
1	컴퓨터소프트웨어공학과
2	컴퓨터정보공학과
3	정보통신과

Navigator

SCHEMAS

Filter objects

- testdb
 - Tables
 - Views
 - Stored Procedures
 - Functions

Administration Schemas

Information

Schema: **testdb**

학과 - Table x

Table Name: **학과** Schema: **testdb**

Charset/Collation: Default Charset Default Collation Engine: InnoDB

Comments:

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
학과번호	INT	☒	☒	☐	☐	☐	☐	☐	☐	
학과명	VARCHAR(50)	☐	☐	☐	☐	☐	☐	☐	☐	

Column Name: Data Type:

Charset/Collation: Default Charset Default Collation Default:

Comments:

Storage: ☐ Virtual ☐ Stored

☐ Primary Key ☐ Not Null ☐ Unique

☐ Binary ☐ Unsigned ☐ Zero Fill

☐ Auto Increment ☐ Generated

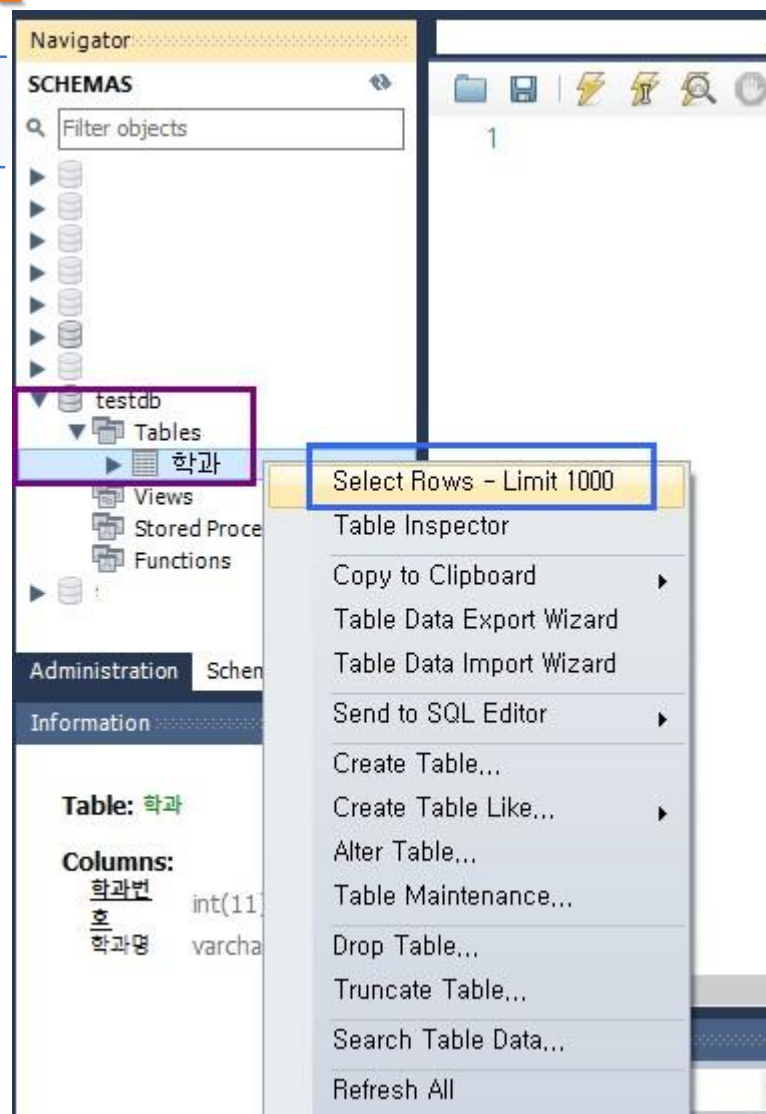
Columns Indexes Foreign Keys Triggers Partitioning Options

Apply **Revert**

[실습] MySQL

3단계: 데이터 입력

학과



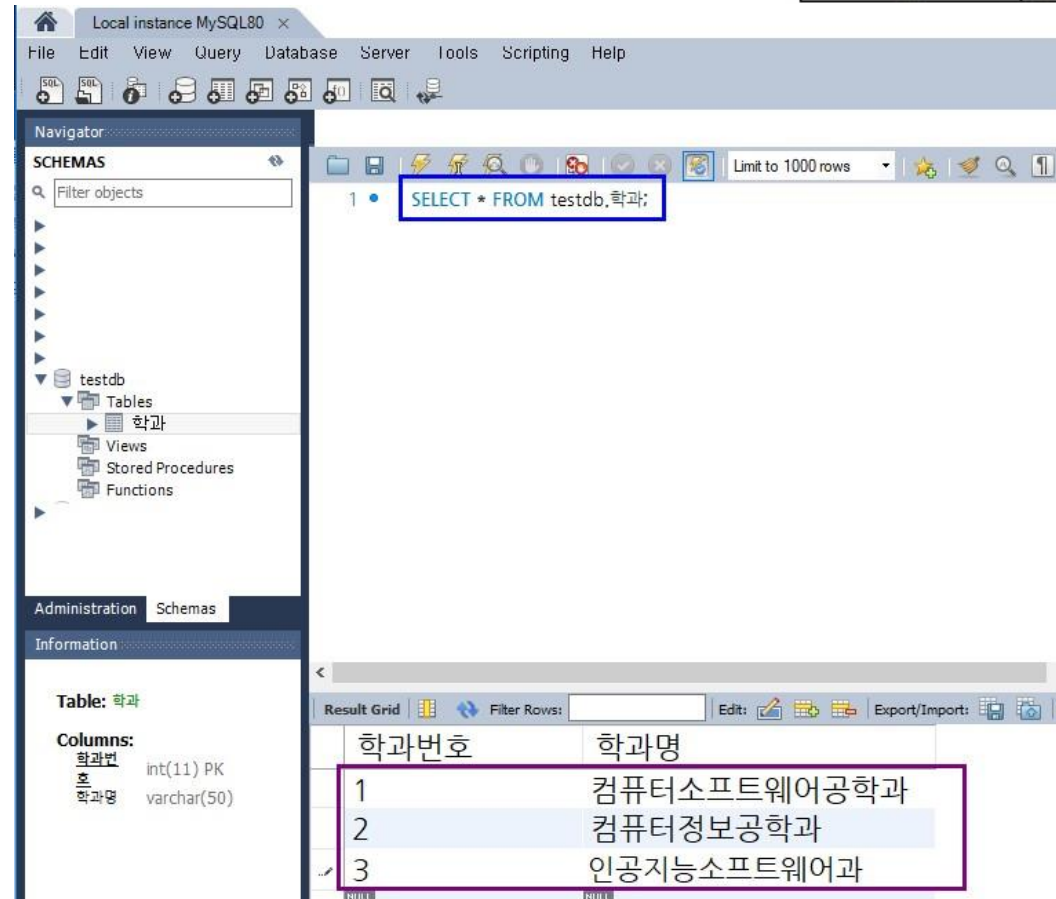
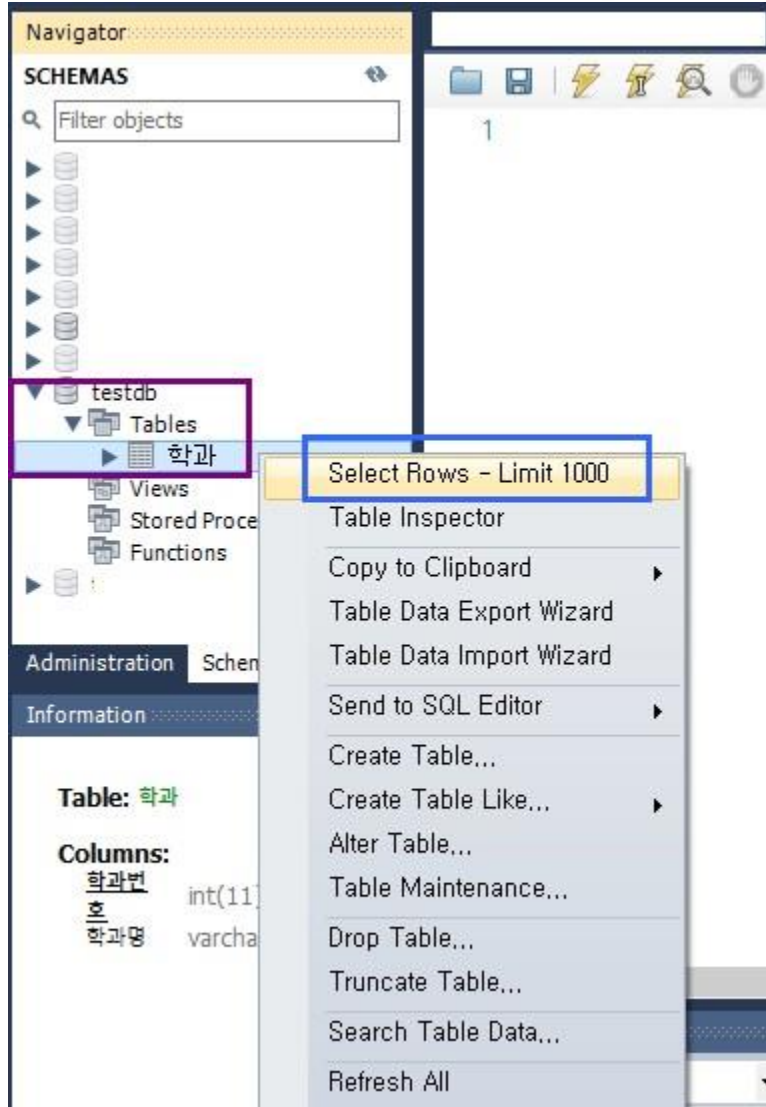
학과

학과번호	학과명
1	컴퓨터소프트웨어공학과
2	컴퓨터정보공학과
3	정보통신과

[실습] MySQL

학과

학과번호	학과명
1	컴퓨터소프트웨어공학과
2	컴퓨터정보공학과
3	정보통신과



[실습] MySQL

학과

학과번호	학과명
1	컴퓨터소프트웨어공학과
2	컴퓨터정보공학과
3	정보통신과

3단계:

MySQL Workbench interface showing a query result for the '학과' table in the 'testdb' database. The result grid displays the following data:

학과번호	학과명
1	컴퓨터소프트웨어공학과
2	컴퓨터정보공학과
3	인공지능소프트웨어과

데이터베이스 개념

2주차

담당교수: 김희숙
(jasmin11@hanmail.net)

데이터베이스



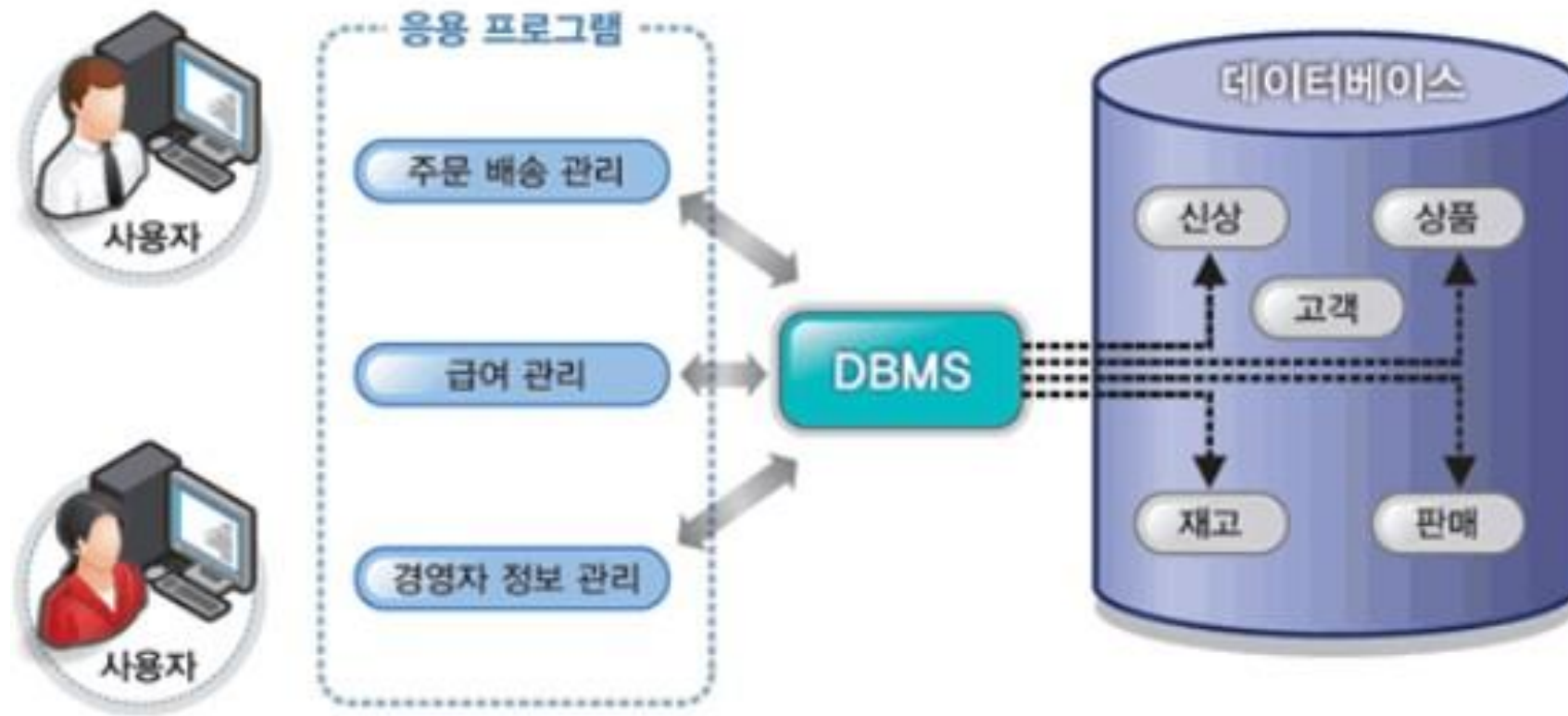
- **데이터베이스:**

- 관련된 데이터 모임
- 한 조직의 여러 응용시스템들이 데이터를 **공유**할 수 있도록 **통합**, **저장**된 **운영**데이터의 집합체

예)

학사 데이터베이스(학생 관련 데이터)
은행 데이터베이스(계좌 거래내역, 신용카드 거래내역 등)
건강보험 데이터베이스(병원의 진료기록 등)
도서관 데이터베이스(도서관 문헌정보 데이터 등)

데이터베이스 구성요소



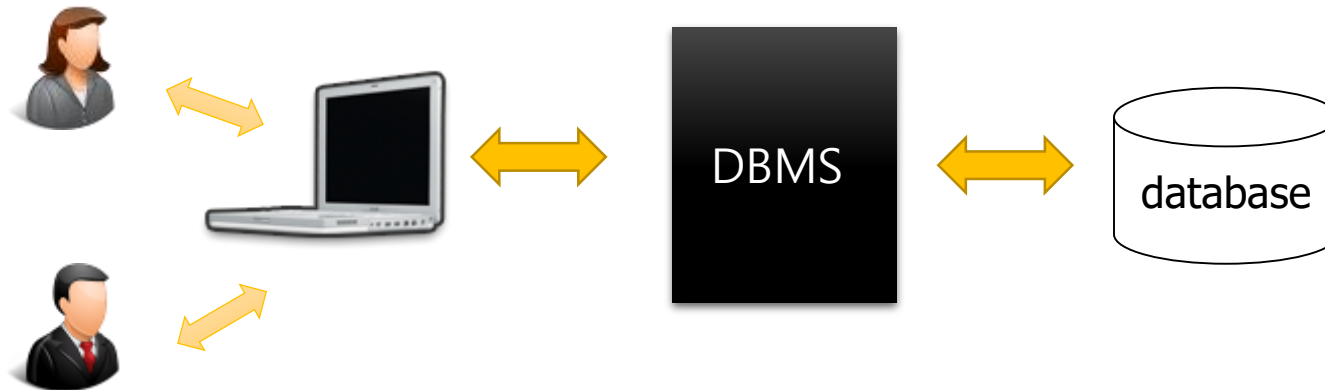
(그림 출처: "Understanding of Database", 이상구외 공저, 이한, 2012)

데이터베이스 관리 시스템(DBMS)

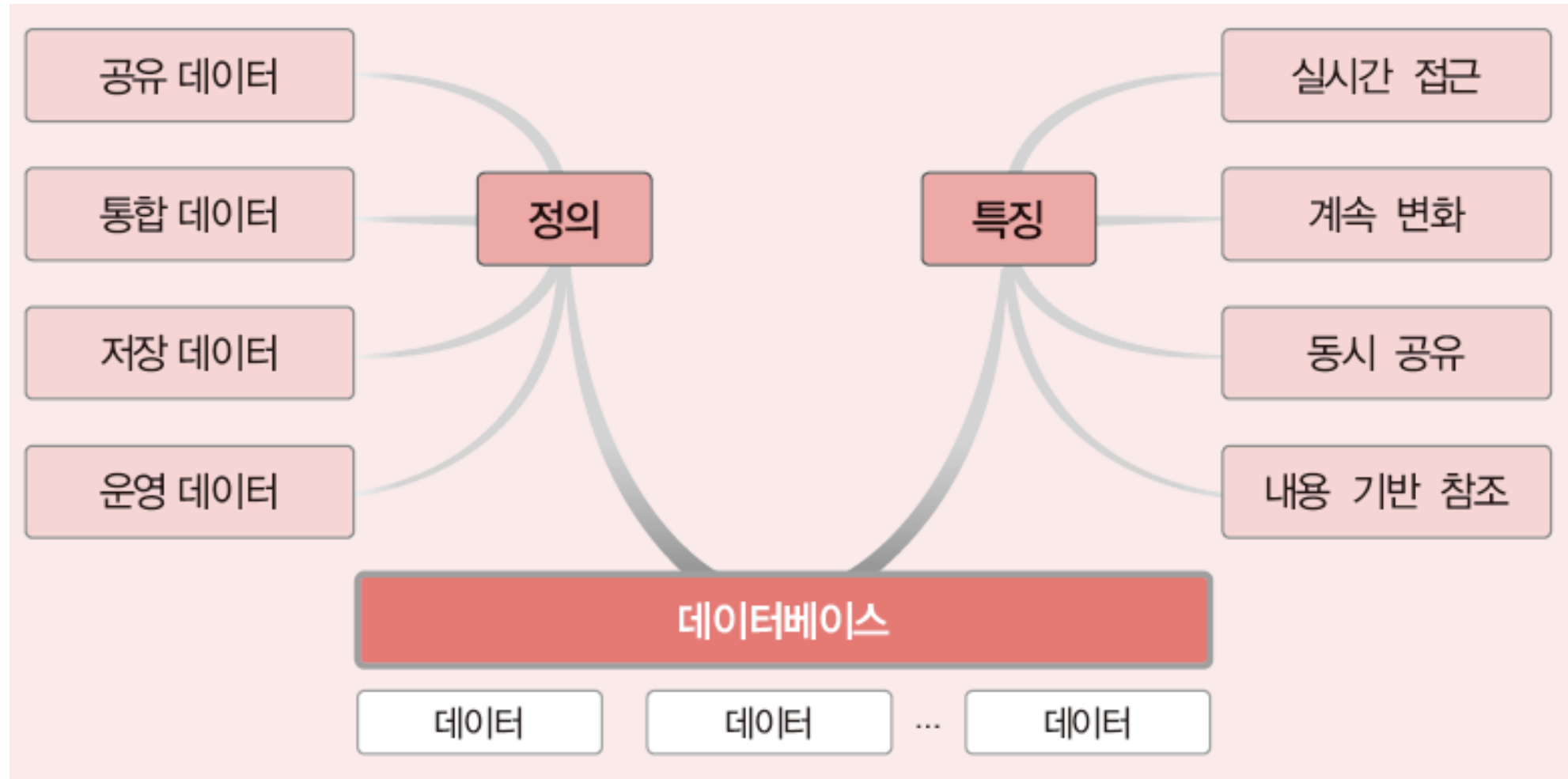


□ 데이터베이스 관리 시스템(DBMS)

- ✓ 데이터베이스를 관리하는 시스템 소프트웨어
 - ✓ MS SQL, MySQL, Oracle, DB2
 - ✓ MariaDB, CUBRID, PostgreSQL



데이터베이스 정의와 특징



(그림 출처: "데이터베이스 개론 2판", 김연희 저, 한빛아카데미, 2019)

실습환경

2주차

담당교수: 김희숙
(jasmin11@hanmail.net)

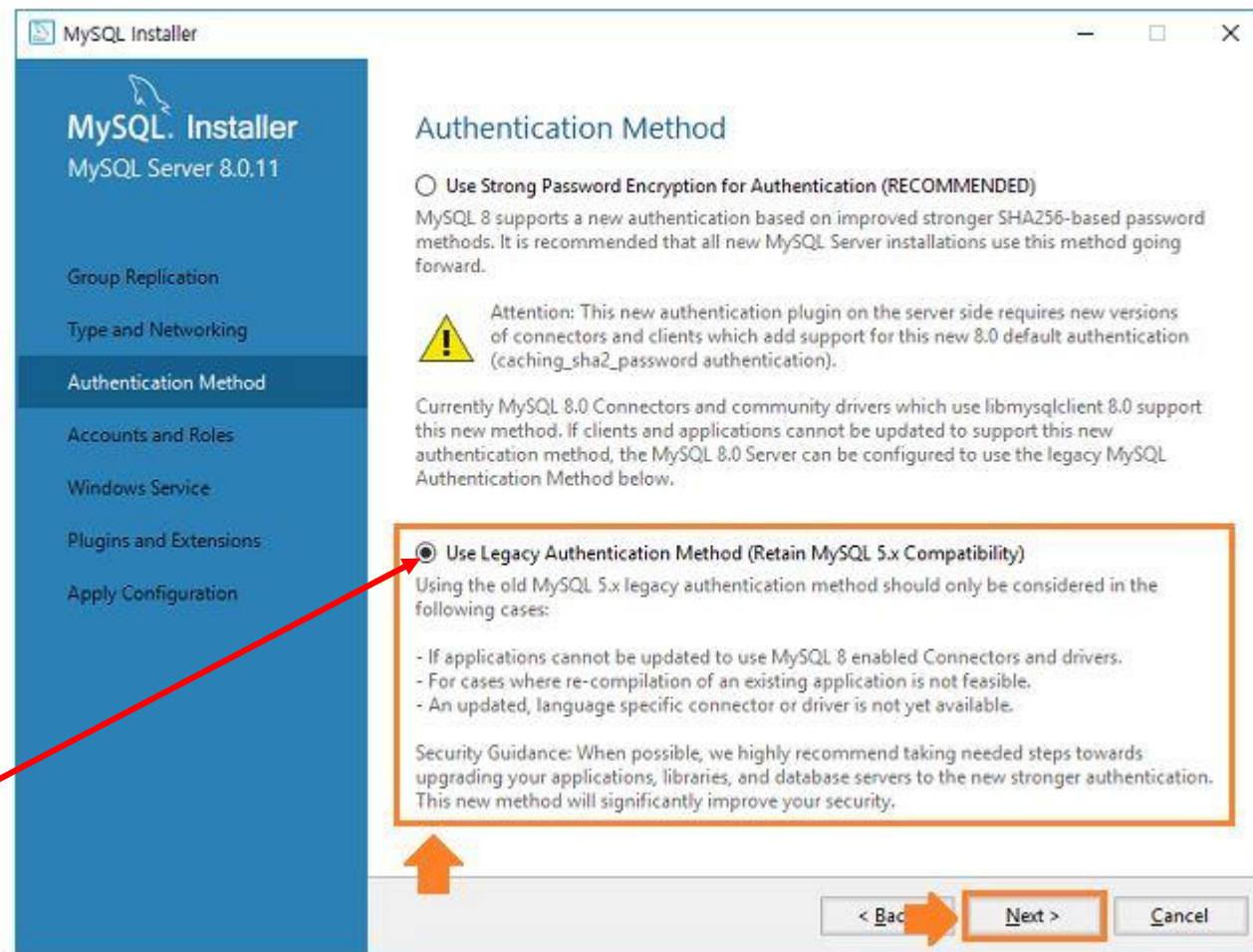
실습 프로그램

□ 프로그램: MySQL8.0

- ✓ MySQL 다운로드 (MySQL Community Server 8.0.21)
- ✓ MySQL Community (GPL) Downloads
- ✓ <http://www.mysql.com/downloads/>
- ✓ <https://dev.mysql.com/downloads/mysql/>

□ 설치: MySQL8.0 (설치할 때, 주의사항 준수)

- ✓ 참고) **MySQL8.0** 설치 경우
하위버전(**MySQL5.x**)과 호환하고자 할 때,
- ✓ 오른쪽 화면이 나오면
두번째 버튼(**Use Legacy...**)을 선택한다



[설치] (MySQL)

1. MySQL 다운로드 (MySQL Community Server 8.0.30)

- 1) <https://dev.mysql.com/downloads/mysql/>
- 2) 다운로드 : MySQL Community Server
(mysql-installer-web-community-8.0.30.msi)
- 3) "No thanks, just start my download" 를 클릭하면 로그인 하지 않고
다운로드 가능

MySQL Community Server

mysql.com/downloads/

MySQL Enterprise Edition

MySQL Enterprise Edition includes the most comprehensive set of advanced features and management tools for MySQL.

- MySQL Database
- MySQL Storage Engines (InnoDB, MyISAM, etc.)
- MySQL Connectors (JDBC, ODBC, .Net, etc.)
- MySQL Replication
- MySQL Partitioning
- MySQL Router
- MySQL Shell
- MySQL Workbench
- 24x7 Technical Support
- MySQL Enterprise Backup
- MySQL Enterprise Monitor
- MySQL Enterprise HA
- MySQL Enterprise Transparent Data Encryption (TDE)
- MySQL Enterprise Masking and De-identification
- MySQL Enterprise Firewall
- MySQL Enterprise Encryption
- MySQL Enterprise Audit

[Learn More](#)[Customer Download](#)[Trial Download](#)

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Japan: 0120-065556
China: 10800-811-0823
India: 0008001005870

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Oracle MySQL Cloud Service

Oracle MySQL Cloud Service is built on MySQL Enterprise Edition and powered by Oracle Cloud, providing an enterprise-grade MySQL database service.

[Learn More »](#)

MySQL Cluster CGE

MySQL Cluster is a real-time open source transactional database designed for fast, always-on access to data under high throughput conditions.

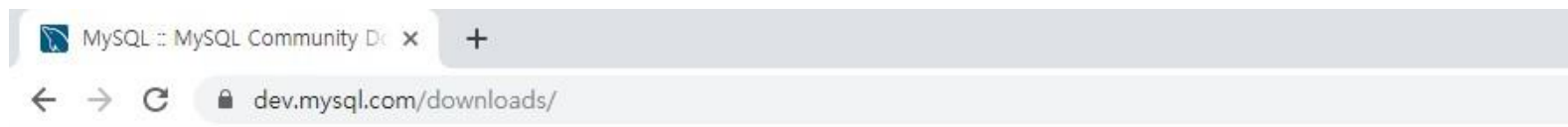
- MySQL Cluster
- MySQL Cluster Manager
- Plus, everything in MySQL Enterprise Edition

[Learn More »](#)

[Customer Download »](#) (Select Patches & Updates Tab, Product Search)

[Trial Download »](#)

[MySQL Community \(GPL\) Downloads »](#)



MySQL Community Downloads

- MySQL Yum Repository
- MySQL APT Repository
- MySQL SUSE Repository
- **MySQL Community Server**
- MySQL Cluster
- MySQL Router
- MySQL Shell
- MySQL Workbench
- MySQL Installer for Windows
- MySQL for Excel
- MySQL for Visual Studio
- MySQL Notifier
- Connector/C (libmysqlclient)
- Connector/C++
- Connector/J
- Connector/NET
- Connector/Node.js
- Connector/ODBC
- Connector/Python
- MySQL Native Driver for PHP
- MySQL Benchmark Tool
- Time zone description tables
- Download Archives



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MySQL Community Downloads

MySQL Community Server

Generally Available (GA) Releases

MySQL Community Server 8.0.17

Select Operating System:

Microsoft Windows

Looking for previous GA versions?

Recommended Download:

MySQL Installer for Windows

All MySQL Products. For All Windows Platforms.
In One Package.

Starting with MySQL 5.6 the MySQL Installer package replaces the standalone MSI packages.

Windows (x86, 32 & 64-bit), MySQL Installer MSI

Go to Download Page >

Other Downloads:

Windows (x86, 64-bit), ZIP Archive

8.0.17

254.2M

Download

(mysql-8.0.17-winx64.zip)

MD5: d120bb0513c2ccfaeee74b0e99217bb7 | Signature

Windows (x86, 64-bit), ZIP Archive

8.0.17

388.2M

Download

Debug Binaries & Test Suite

(mysql-8.0.17-winx64-debug-test.zip)

MD5: 16478256aeda23d9df35406fa6f504d3 | Signature



We suggest that you use the MD5 checksums and GnuPG signatures to verify the integrity of the packages you download.

← → ↻ dev.mysql.com/downloads/windows/installer/8.0.html

MySQL Community Downloads

MySQL Installer

Generally Available (GA) Releases

MySQL Installer 8.0.17

Select Operating System:

Microsoft Windows ▼

[Looking for previous GA versions?](#)

Windows (x86, 32-bit), MSI Installer

8.0.17

18.5M

[Download](#)

(mysql-installer-web-community-8.0.17.0.msi)

MD5: 567707887fc0d1fad7fc848a878a0da2 | [Signature](#)

Windows (x86, 32-bit), MSI Installer

8.0.17

393.4M

[Download](#)

(mysql-installer-community-8.0.17.0.msi)

MD5: 3aa8d6470fb6b58f517d3efb46e5472b | [Signature](#)



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MySQL Community Downloads

Login Now or Sign Up for a free account.

An Oracle Web Account provides you with the following advantages:

- Fast access to MySQL software downloads
- Download technical White Papers and Presentations
- Post messages in the MySQL Discussion Forums
- Report and track bugs in the MySQL bug system

Login »

using my Oracle Web account

Sign Up »

for an Oracle Web account

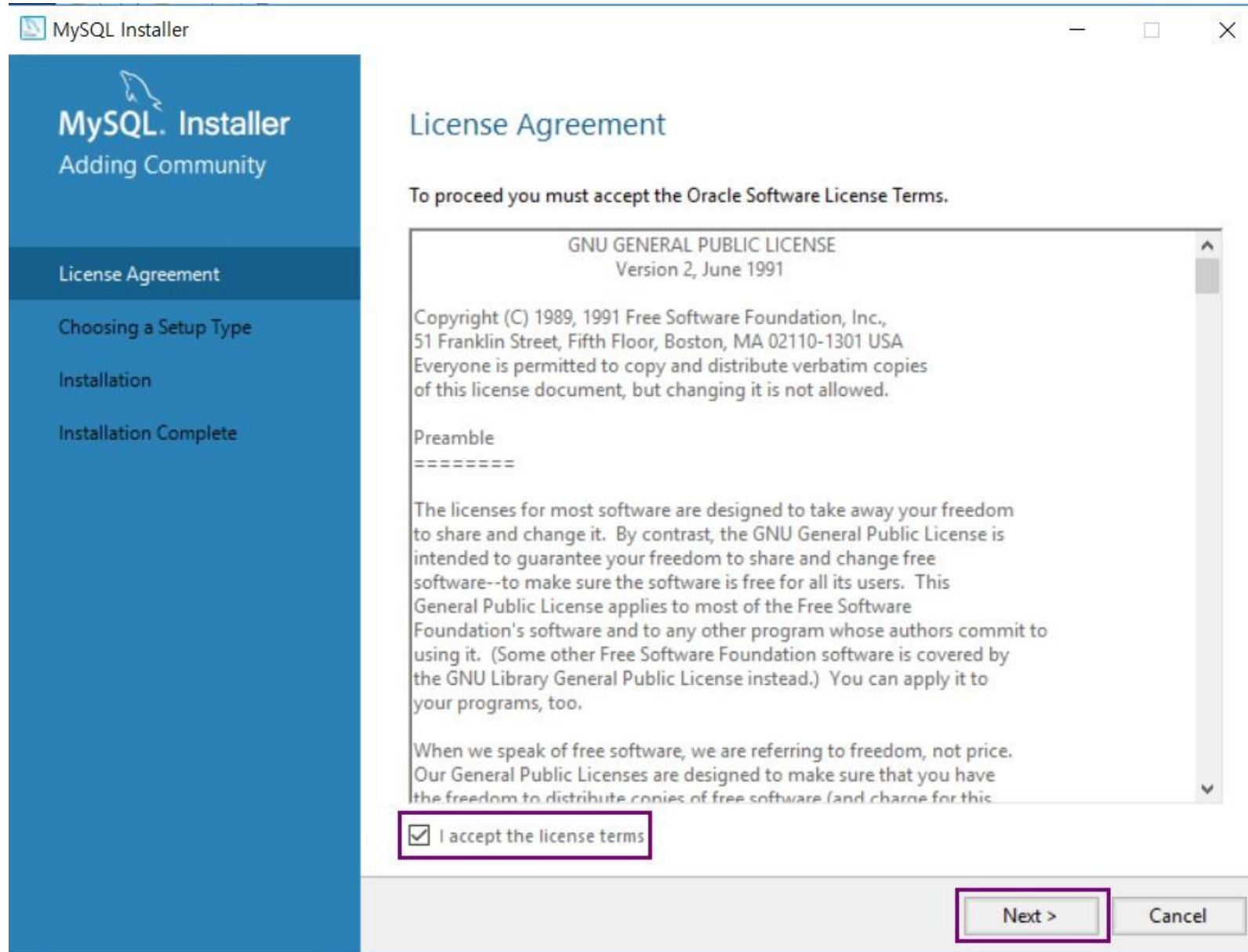
MySQL.com is using Oracle SSO for authentication. If you already have an Oracle Web account, click the Login link. Otherwise, you can signup for a free account by clicking the Sign Up link and following the instructions.

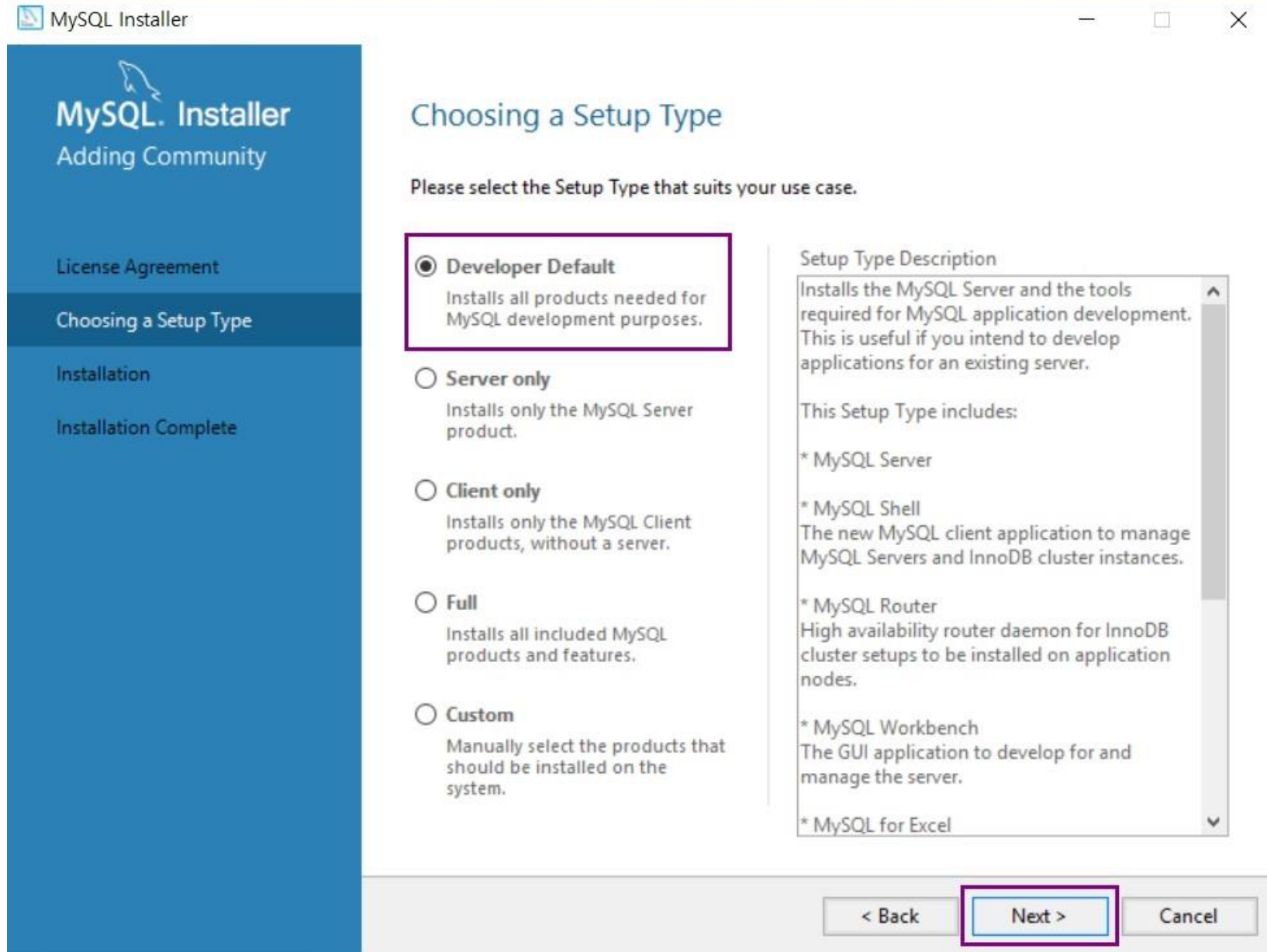
No thanks, just start my download.

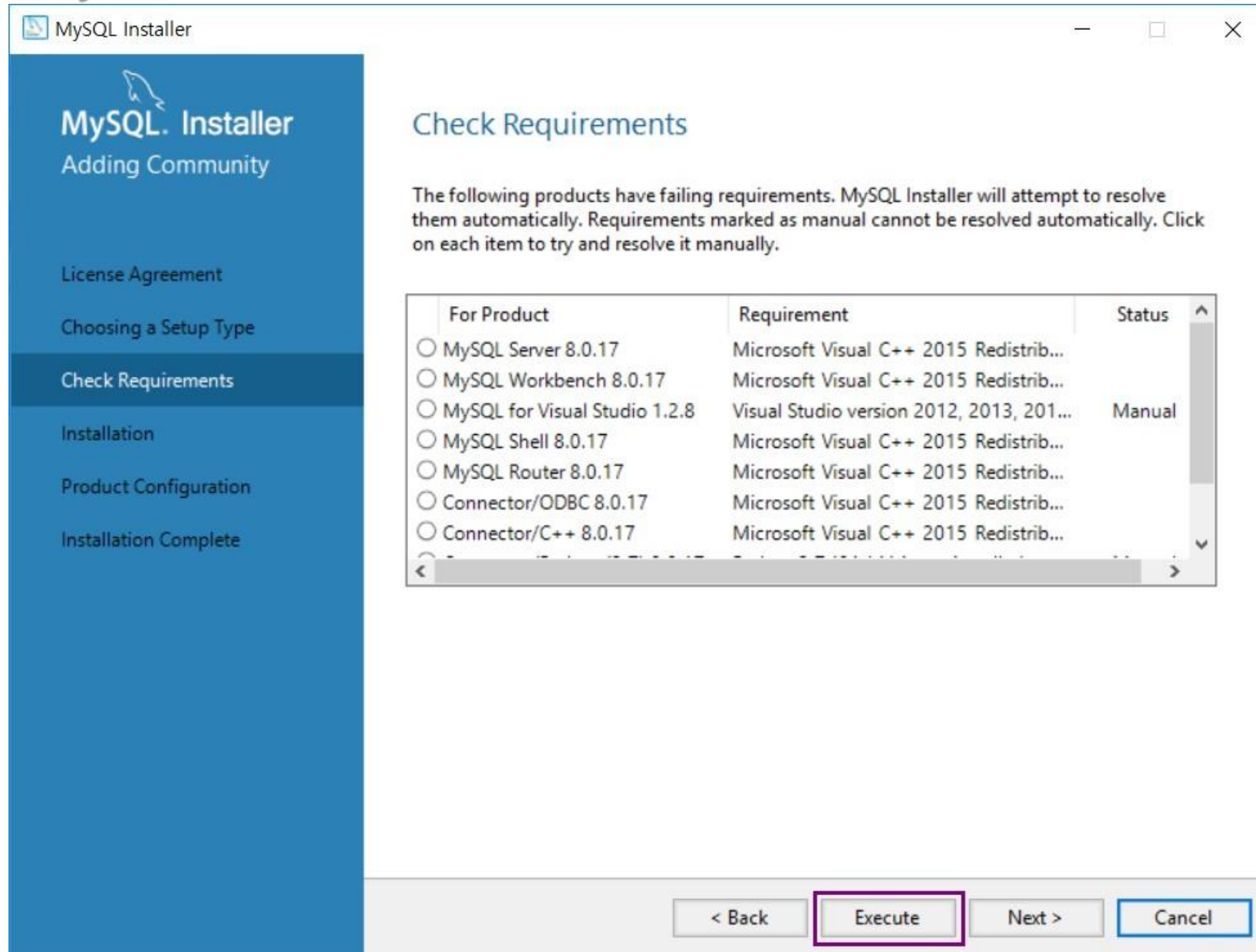
ORACLE

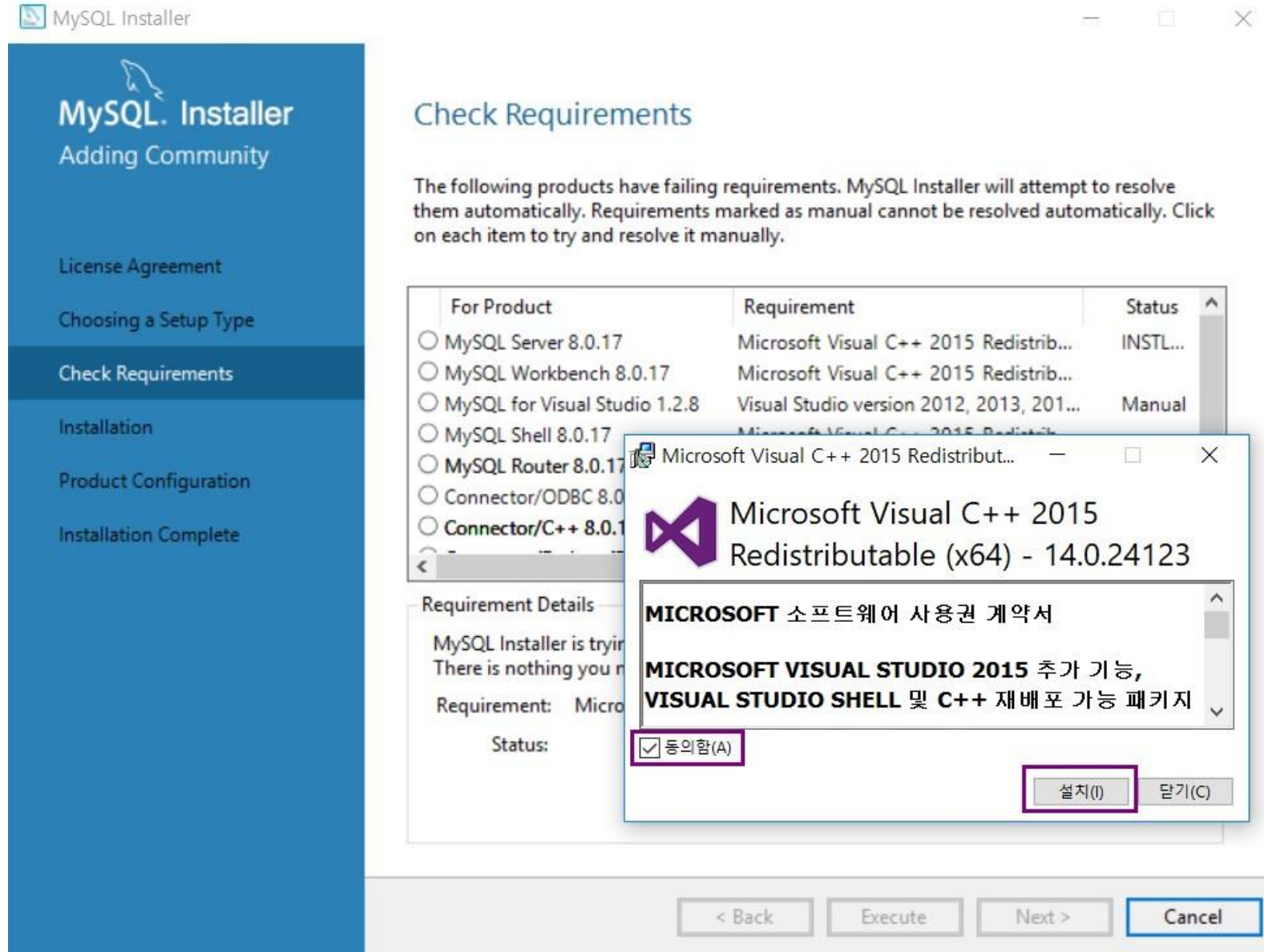
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MySQL Installer

MySQL. Installer Adding Community

License Agreement

Choosing a Setup Type

Check Requirements

Installation

Product Configuration

Installation Complete

Check Requirements

The following products have failing requirements. MySQL Installer will attempt to resolve them automatically. Requirements marked as manual cannot be resolved automatically. Click on each item to try and resolve it manually.

For Product	Requirement	Status
☒ MySQL Server 8.0.17	Microsoft Visual C++ 2015 Redistrib...	INSTL DON
☒ MySQL Workbench 8.0.17	Microsoft Visual C++ 2015 Redistrib...	INSTL DON
☐ MySQL for Visual Studio 1.2.8	Visual Studio version 2012, 2013, 201...	Manual
☒ MySQL Shell 8.0.17	Microsoft Visual C++ 2015 Redistrib...	INSTL DON
☒ MySQL Router 8.0.17	Microsoft Visual C++ 2015 Redistrib...	INSTL DON
☒ Connector/ODBC 8.0.17	Microsoft Visual C++ 2015 Redistrib...	INSTL DON
☒ Connector/C++ 8.0.17	Microsoft Visual C++ 2015 Redistrib...	INSTL DON

Requirement Details

MySQL Installer is trying to automatically resolve this requirement.
There is nothing you need to do.

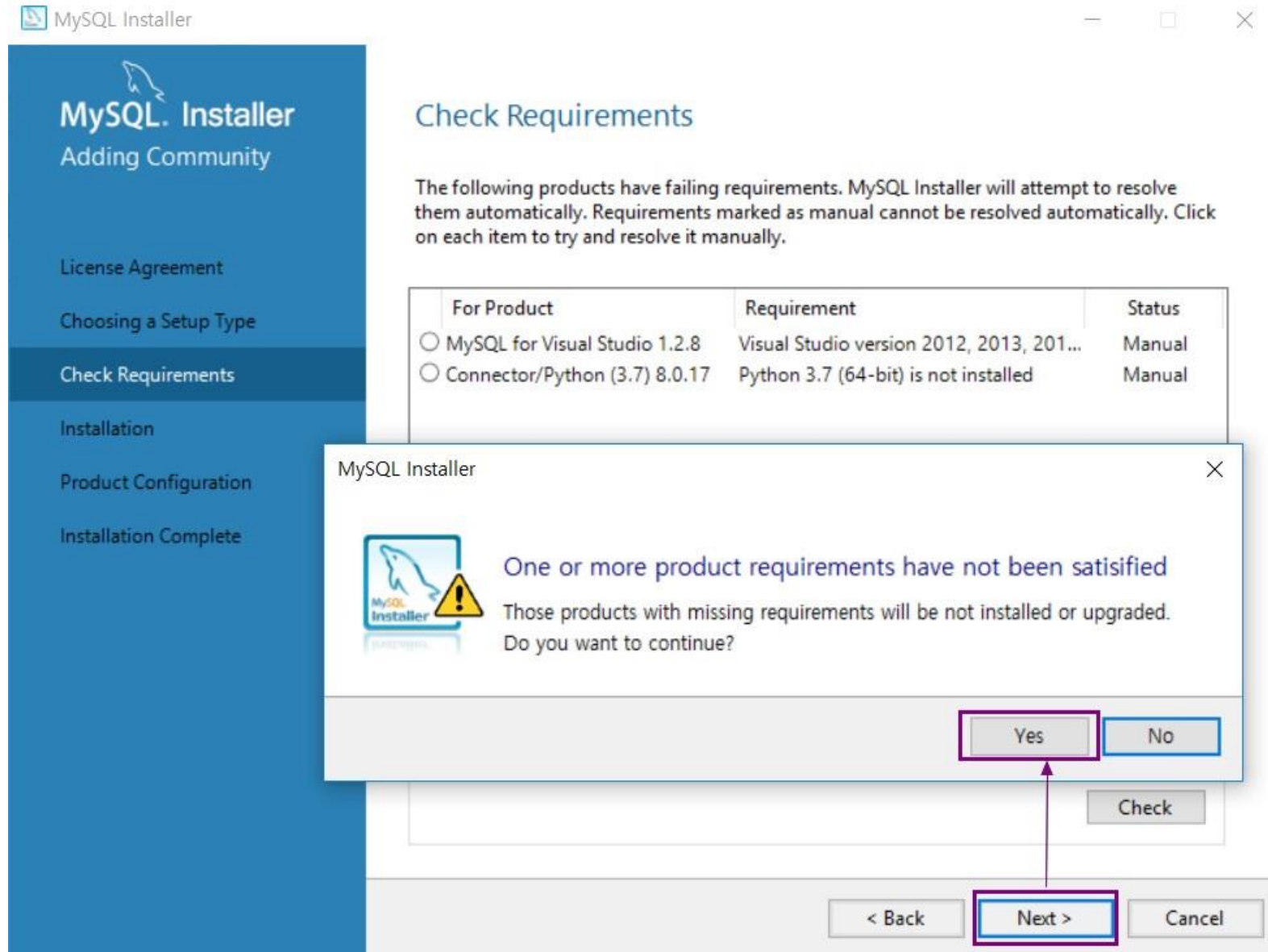
Requirement: Microsoft Visual C++ 2015 Redistributable Package (x64) is not installed

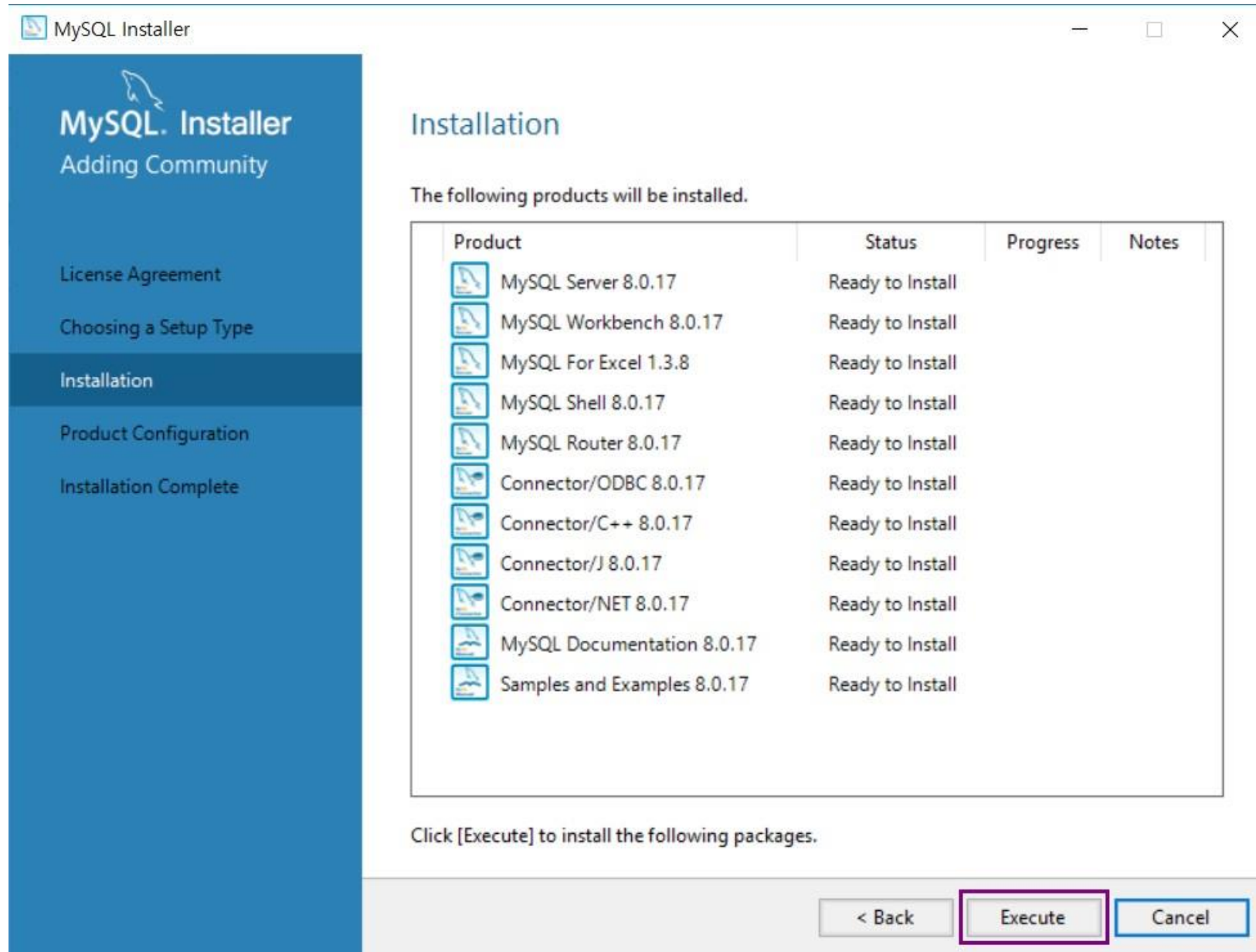
Status:

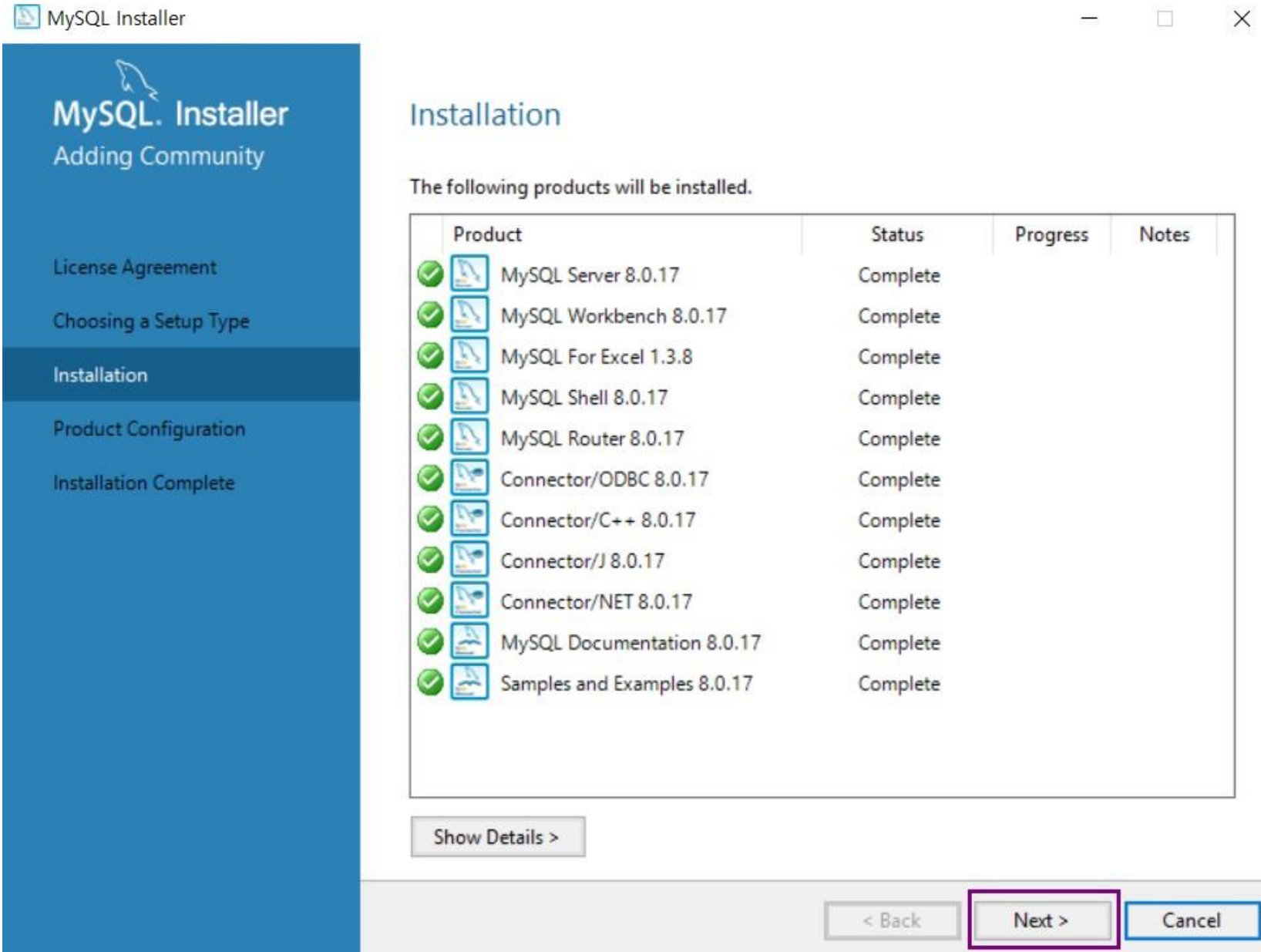
< Back

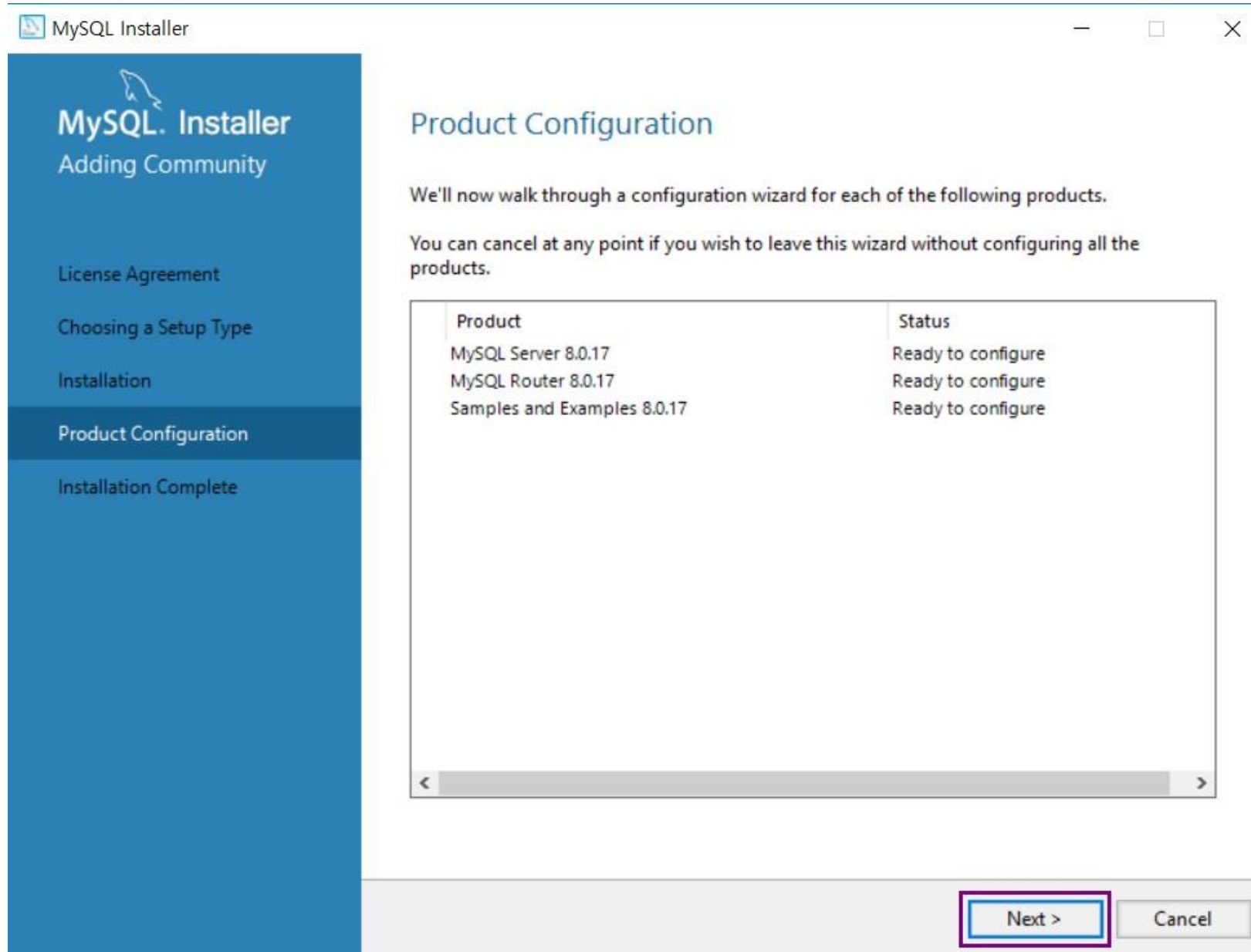
Next >

Cancel









MySQL Installer

MySQL. Installer MySQL Server 8.0.17

High Availability

Type and Networking

Authentication Method

Accounts and Roles

Windows Service

Logging Options

Advanced Options

Apply Configuration

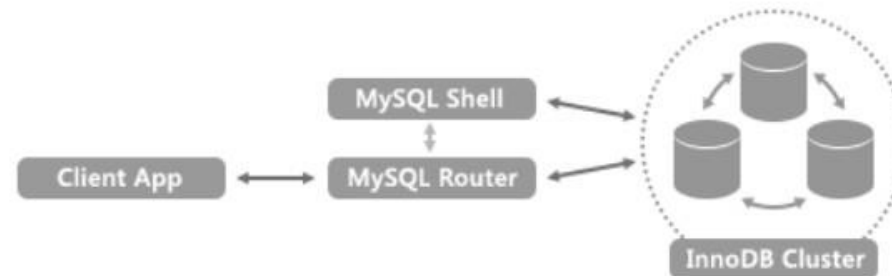
High Availability

☒ Standalone MySQL Server / Classic MySQL Replication

Choose this option to run the MySQL instance as a standalone database server with the opportunity to configure classic replication later. With this option, you can provide your own high-availability solution, if required.

☐ InnoDB Cluster

The InnoDB cluster technology provides an out-of-the-box high availability (HA) solution for MySQL using Group Replication.



Note: [InnoDB cluster](#) requires a minimum of three MySQL server instances to provide a fully automated HA solution. Members of a cluster should be located such that network communication latency between servers is low.

Next >

Cancel

MySQL Installer

MySQL Server 8.0.17

High Availability

Type and Networking

Authentication Method

Accounts and Roles

Windows Service

Apply Configuration

Type and Networking

Server Configuration Type

Choose the correct server configuration type for this MySQL Server installation. This setting will define how much system resources are assigned to the MySQL Server instance.

Config Type: Development Computer

Connectivity

Use the following controls to select how you would like to connect to this server.

☒ TCP/IP Port: 3306 X Protocol Port: 33060

☒ Open Windows Firewall ports for network access

☐ Named Pipe Pipe Name: MYSQL

☐ Shared Memory Memory Name: MYSQL

Advanced Configuration

Select the check box below to get additional configuration pages where you can set advanced and logging options for this server instance.

☐ Show Advanced and Logging Options

< Back Next > Cancel



MySQL Installer

MySQL Installer MySQL Server 8.0.17

High Availability

Type and Networking

Authentication Method

Accounts and Roles

Windows Service

Apply Configuration

Authentication Method

☐ Use Strong Password Encryption for Authentication (RECOMMENDED)

MySQL 8 supports a new authentication based on improved stronger SHA256-based password methods. It is recommended that all new MySQL Server installations use this method going forward.



Attention: This new authentication plugin on the server side requires new versions of connectors and clients which add support for this new 8.0 default authentication (caching_sha2_password authentication).

Currently MySQL 8.0 Connectors and community drivers which use libmysqlclient 8.0 support this new method. If clients and applications cannot be updated to support this new authentication method, the MySQL 8.0 Server can be configured to use the legacy MySQL Authentication Method below.

☒ Use Legacy Authentication Method (Retain MySQL 5.x Compatibility)

Using the old MySQL 5.x legacy authentication method should only be considered in the following cases:

- If applications cannot be updated to use MySQL 8 enabled Connectors and drivers.
- For cases where re-compilation of an existing application is not feasible.
- An updated, language specific connector or driver is not yet available.

Security Guidance: When possible, we highly recommend taking needed steps towards upgrading your applications, libraries, and database servers to the new stronger authentication. This new method will significantly improve your security.

< Back

Next >

Cancel

MySQL Installer

MySQL. Installer
MySQL Server 8.0.17

- High Availability
- Type and Networking
- Authentication Method
- Accounts and Roles**
- Windows Service
- Apply Configuration

Accounts and Roles

Root Account Password
Enter the password for the root account. Please remember to store this password in a secure place.

MySQL Root Password:

Repeat Password:

root 암호: admin

Password strength: **Weak**

MySQL User Accounts

Create MySQL user accounts for your users and applications. Assign a role to the user that consists of a set of privileges.

MySQL User Name	Host	User Role
-----------------	------	-----------

MySQL Installer

MySQL. Installer
MySQL Server 8.0.17

High Availability

Type and Networking

Authentication Method

Accounts and Roles

Windows Service

Apply Configuration

Windows Service

☒ Configure MySQL Server as a Windows Service

Windows Service Details
Please specify a Windows Service name to be used for this MySQL Server instance. A unique name is required for each instance.

Windows Service Name:

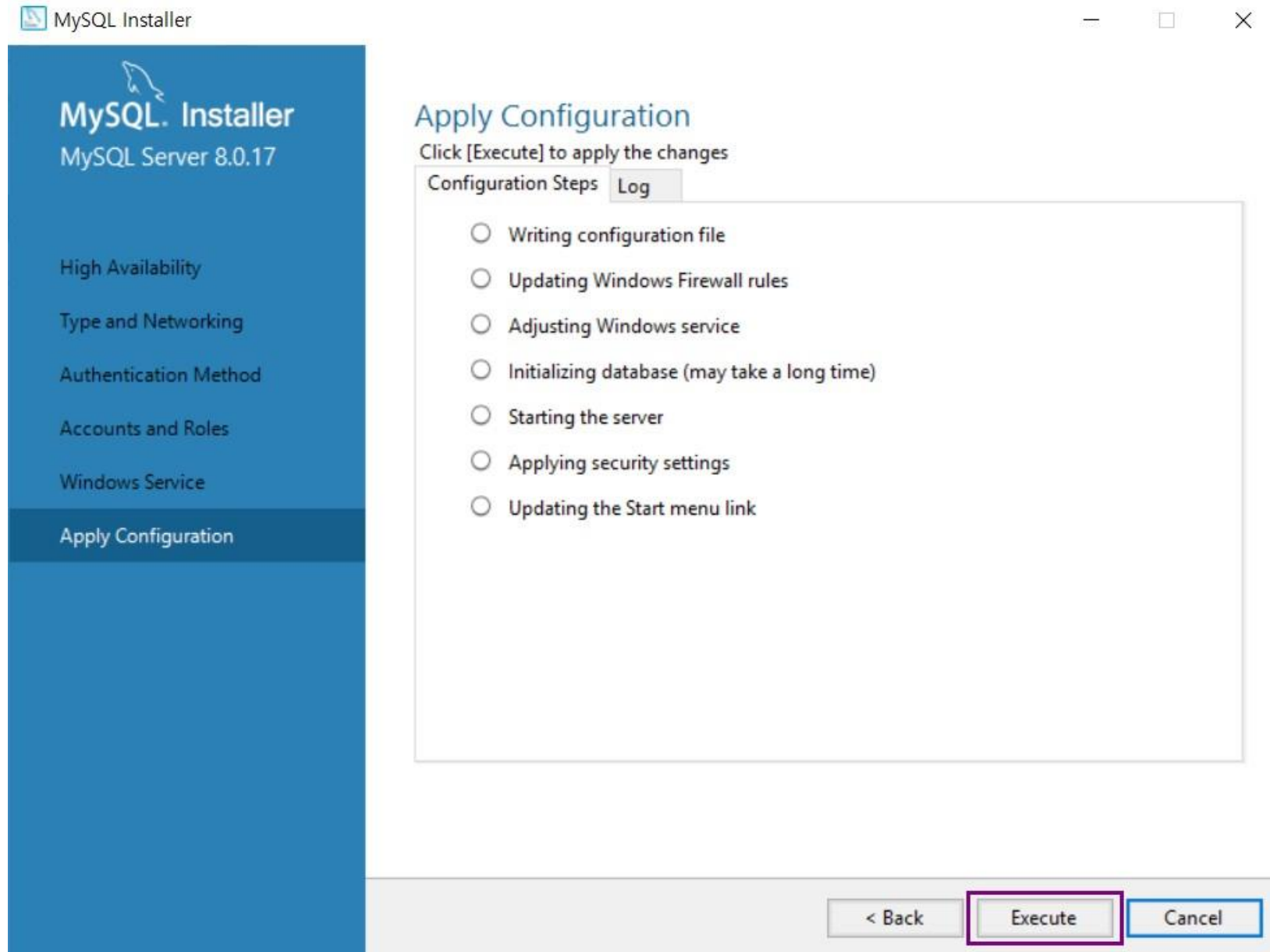
☒ Start the MySQL Server at System Startup

Run Windows Service as ...
The MySQL Server needs to run under a given user account. Based on the security requirements of your system you need to pick one of the options below.

☒ Standard System Account
Recommended for most scenarios.

☐ Custom User
An existing user account can be selected for advanced scenarios.

< Back **Next >** Cancel



MySQL Installer

MySQL Installer MySQL Server 8.0.17

High Availability

Type and Networking

Authentication Method

Accounts and Roles

Windows Service

Apply Configuration

Apply Configuration

The configuration operation has finished.

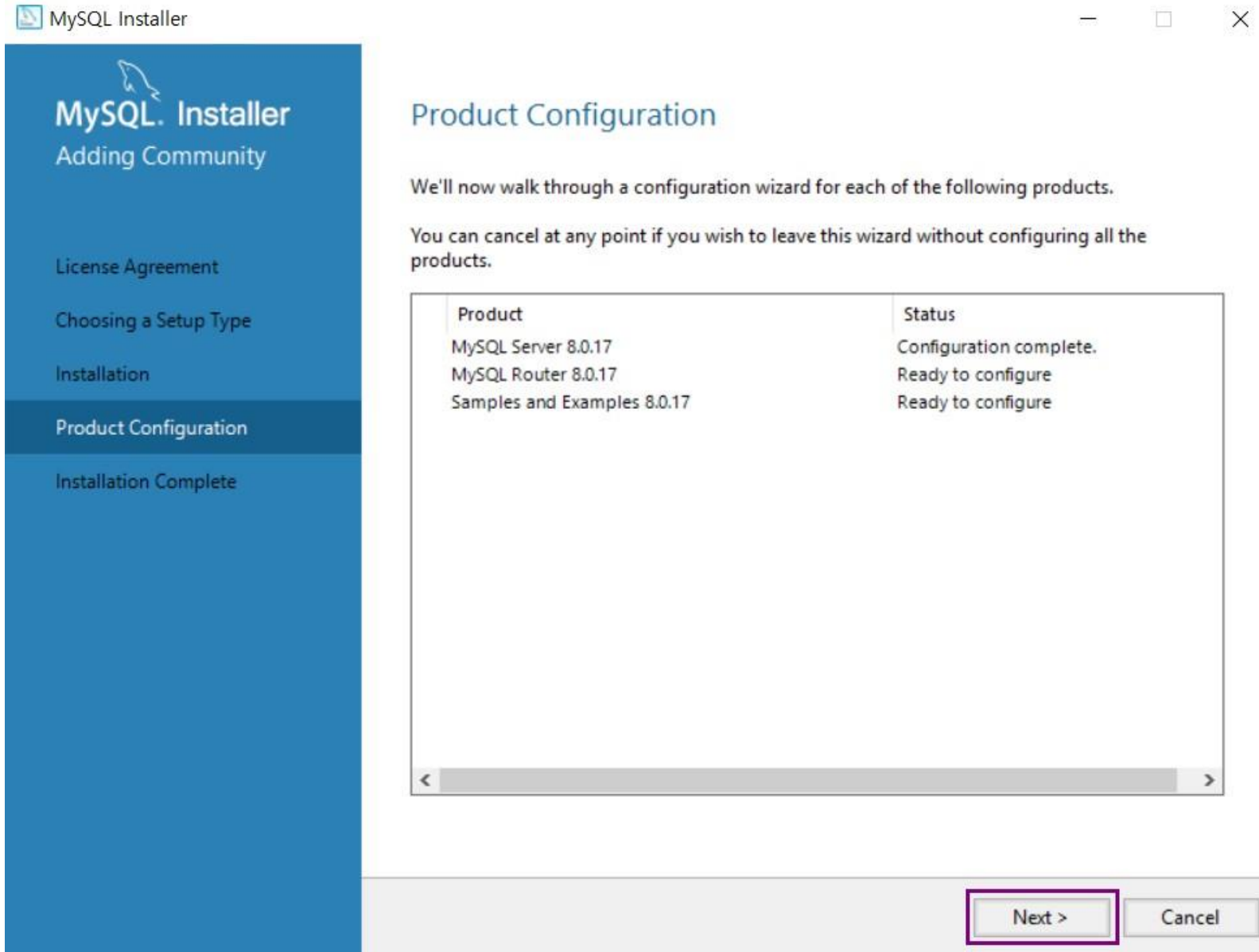
Configuration Steps

Log

- ✓ Writing configuration file
- ✓ Updating Windows Firewall rules
- ✓ Adjusting Windows service
- ✓ Initializing database (may take a long time)
- ✓ Starting the server
- ✓ Applying security settings
- ✓ Updating the Start menu link

The configuration for MySQL Server 8.0.17 was successful.
Click Finish to continue.

Finish



MySQL Installer

MySQL Router 8.0.17

MySQL Router Configuration

☐ Bootstrap MySQL Router for use with InnoDB cluster

This wizard can bootstrap MySQL Router to direct traffic between MySQL applications and a MySQL InnoDB cluster. Applications that connect to the router will be automatically directed to an available read/write or read-only member of the cluster.

The bootstrapping process requires a connection to the InnoDB cluster. In order to register the MySQL Router for monitoring, use the current Read/Write instance of the cluster.

Hostname:

Port:

Management User:

Password:

MySQL Router requires specification of a base port (between 80 and 65532). The first port is used for classic read/write connections. The other ports are computed sequentially after the first port. If any port is indicated to be in use, please change the base port.

Classic MySQL protocol connections to InnoDB cluster:

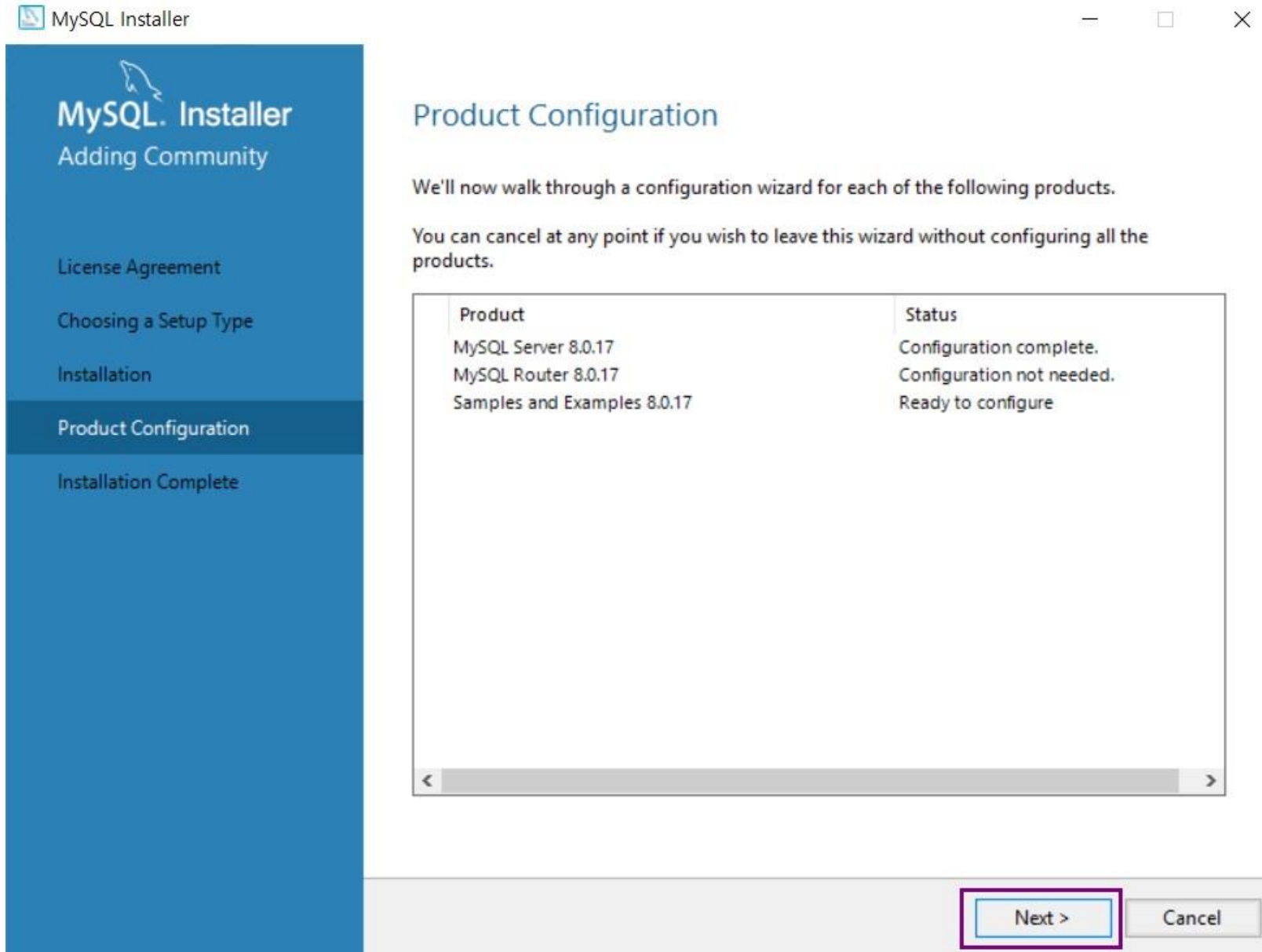
Read/Write:

Read Only:

MySQL X protocol connections to InnoDB cluster:

Read/Write:

Read Only:



MySQL Installer

MySQL. Installer
Samples and Examples

Connect To Server

Apply Configuration

Connect To Server

Select the MySQL server instances from the list to receive sample schemas and data.

☐ Show MySQL Server instances that may be running in read-only mode

	Server	Port	Arch...	Type	Status
☒	MySQL Server 8.0.17	3306	X64	Stand-alone Server	Connection succeeded.

Provide the credentials that should be used (requires root privileges).
Click "Check" to ensure they work.

User name: Credentials provided in Server configuration

Password:



[실습] (설치) MySQL

MySQL Installer

MySQL[®] Installer
Samples and Examples

Connect To Server

Apply Configuration

Apply Configuration

Click [Execute] to apply the changes

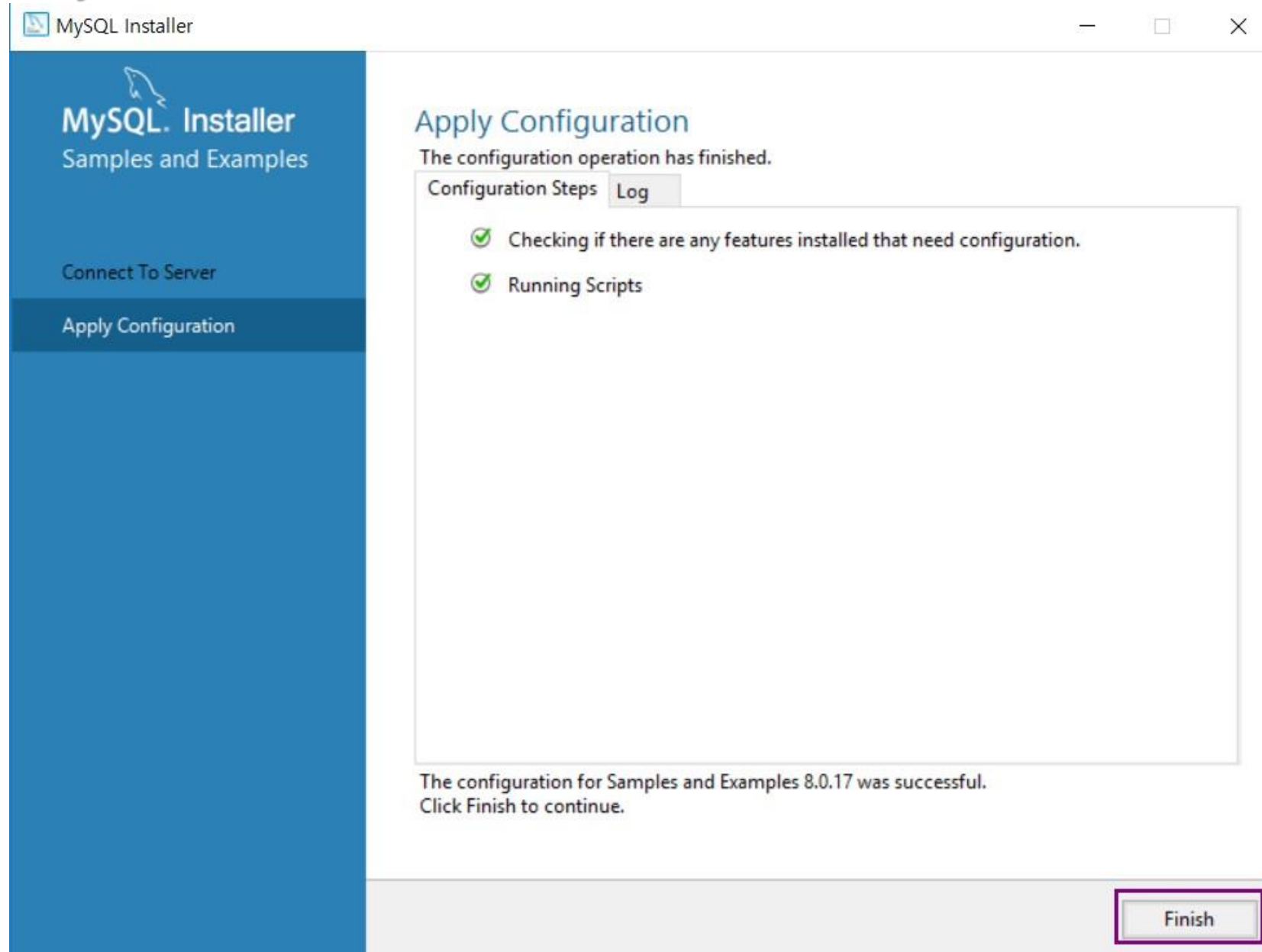
Configuration Steps Log

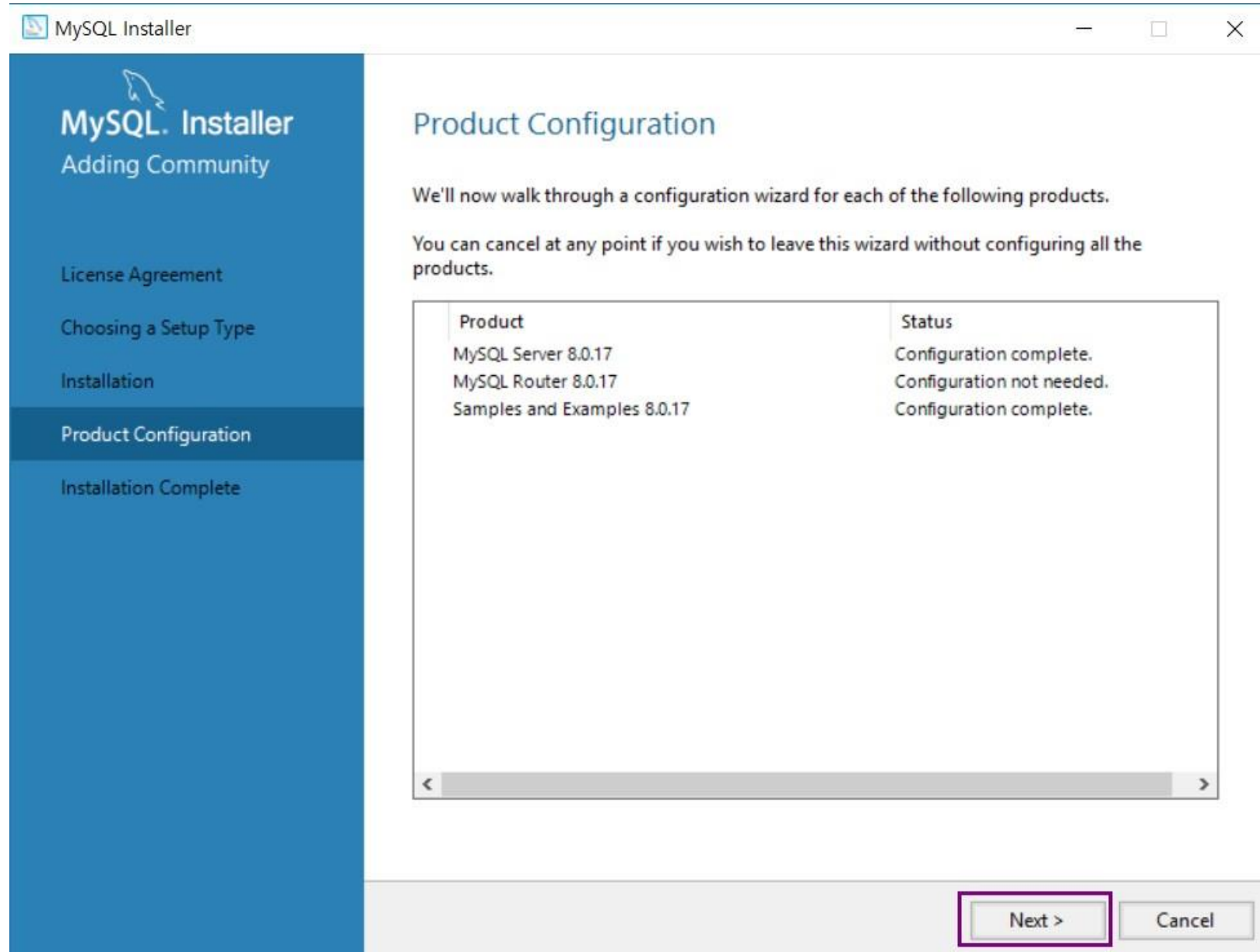
- ☐ Checking if there are any features installed that need configuration.
- ☐ Running Scripts

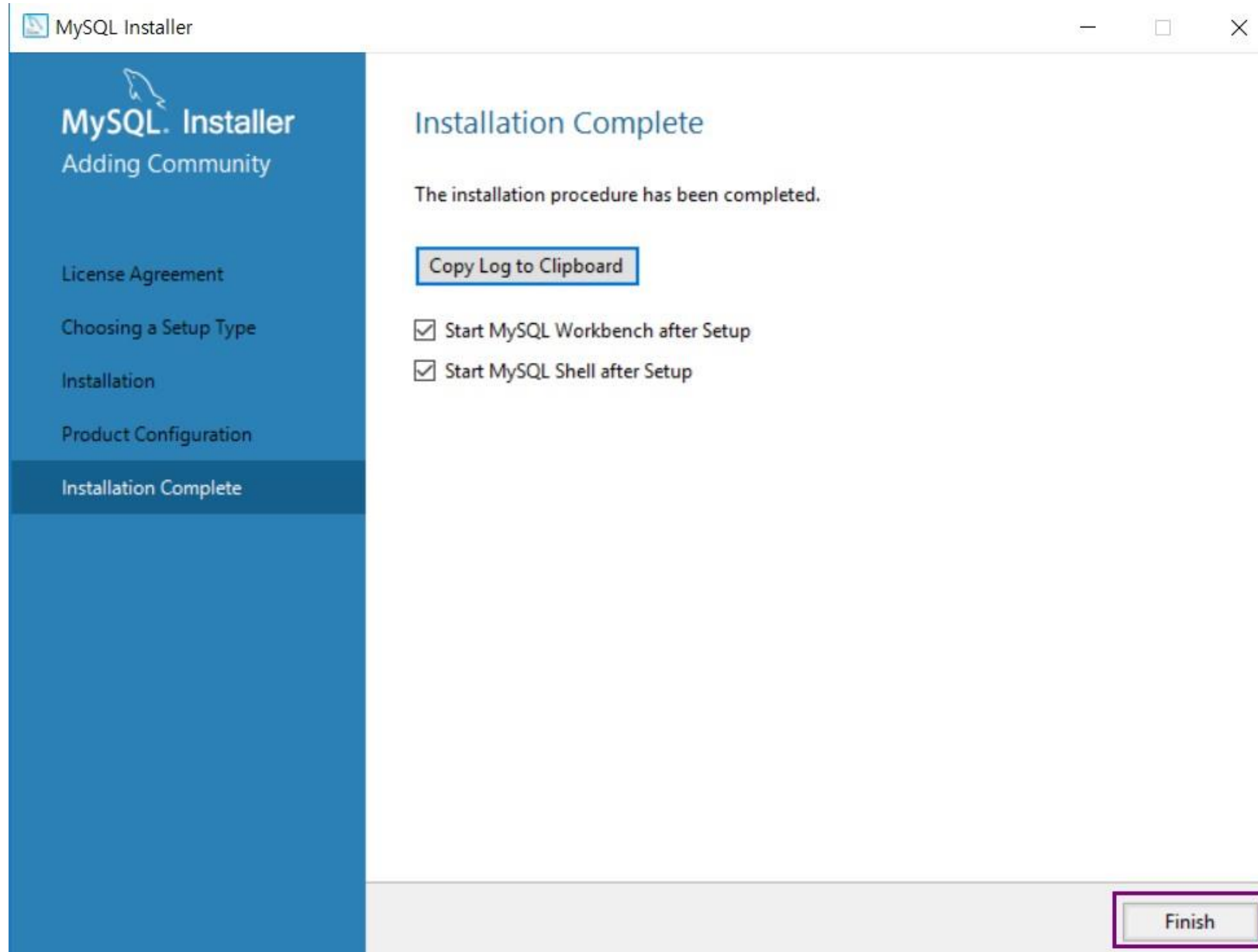
< Back

Execute

Cancel









Welcome to MySQL Workbench

MySQL Workbench is the official graphical user interface (GUI) tool for MySQL. It allows you to design, create and browse your database schemas, work with database objects and insert data as well as design and run SQL queries to work with stored data. You can also migrate schemas and data from other database vendors to your MySQL database.

[Browse Documentation >](#)

[Read the Blog >](#)

[Discuss on the Forums >](#)

MySQL Connections



Local instance MySQL80

 root

 localhost:3306

The screenshot shows the MySQL Workbench 8.0 CE installation window. The 'MySQL Workbench 8.0 CE' option is highlighted with a blue border. Below the installation window, the MySQL Workbench 8.0 CE interface is visible. The main window displays 'Welcome to MySQL Workbench' and a description of the tool. A 'Connect to MySQL Server' dialog box is open, prompting for a password for the 'root' user on 'localhost:3306'. The 'Local instance MySQL80' connection is listed in the 'MySQL Connections' panel.

MySQL Workbench 8.0 CE

MySQL Workbench 8.0 CE

MySQL 8.0 Command Line Client

MySQL 8.0 Command Line Client...

MySQL Installer - Community

MySQL Workbench 8.0 CE

Welcome to MySQL Workbench

MySQL Workbench is the official graphical user interface (GUI) tool for MySQL. It allows you to design, create and browse your database schemas, work with database objects and insert data as well as design and run SQL queries to work with stored data. You can also migrate schemas and data from other

Connect to MySQL Server

Please enter password for the following service:

Service: Mysql@localhost:3306

User: root

Password: *****

☐ Save password in vault

OK Cancel

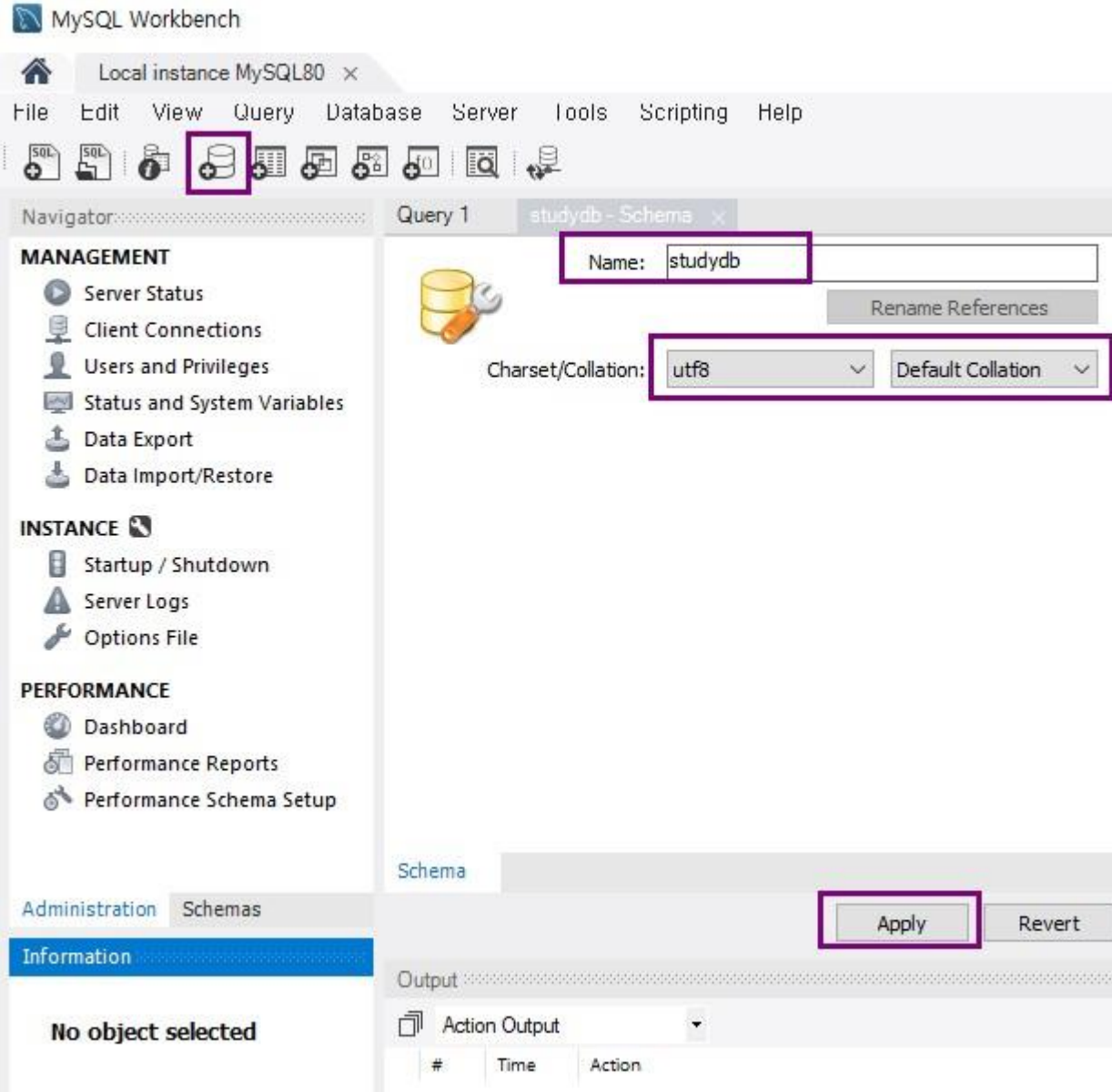
MySQL Connections

Local instance MySQL80

root

localhost:3306

[실습] (실행) MySQL: 스키마 생성



[실습] (실행) MySQL: 스키마 생성

Apply SQL Script to Database

Review SQL Script

Apply SQL Script

Review the SQL Script to be Applied on the Database

Online DDL

Algorithm:

Default

Lock Type:

Default

```
1 CREATE SCHEMA `studydb` DEFAULT CHARACTER SET utf8 ;  
2
```



Back

Apply

Cancel

[실습] (실행) MySQL: 스키마 생성

Apply SQL Script to Database

Review SQL Script
Apply SQL Script

Review the SQL Script to be Applied

Online DDL

Algorithm:

Default

```
1 CREATE SCHEMA `stud
2
```

<

Apply SQL Script to Database

Review SQL Script

Apply SQL Script

Applying SQL script to the database

The following tasks will now be executed. Please monitor the execution.
Press Show Logs to see the execution logs.

☒ Execute SQL Statements

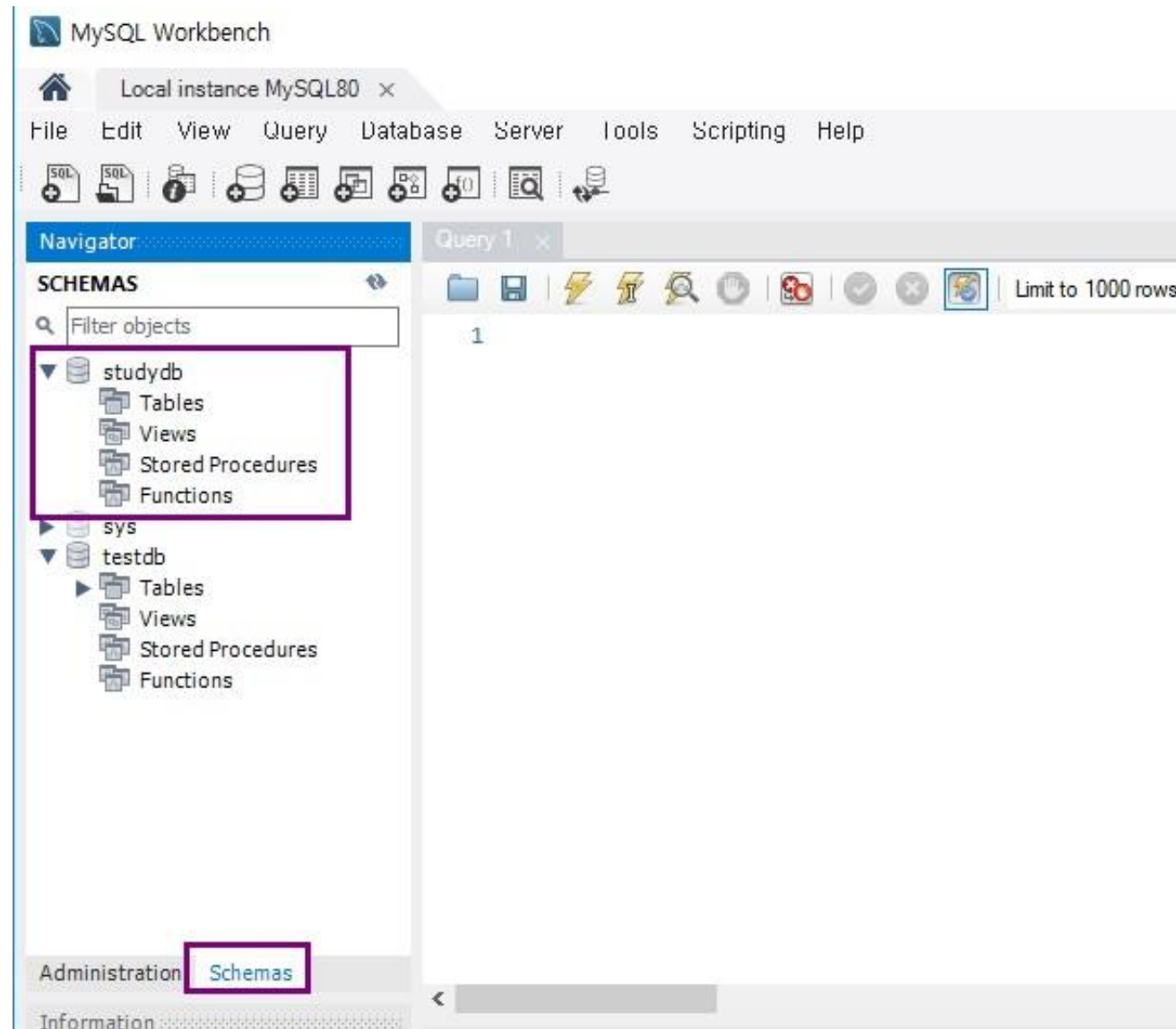
SQL script was successfully applied to the database.

Show Logs

Back

Finish

Cancel



실습

2주차

담당교수: 김희숙
(jasmin11@hanmail.net)

MySQL 실습



-- MySQL Workbench 실행

-- (MySQL)

```
drop database studydb;
```

```
CREATE DATABASE `studydb`  
  DEFAULT CHARACTER SET utf8mb4;
```

```
use studydb;
```

1단계: 스키마/데이터베이스 생성 **studydb**
2단계: 테이블 생성 **학과, 학생**
3단계: 데이터 입력

-- 1단계: 스키마 생성

```
CREATE SCHEMA `studydb`  
  DEFAULT CHARACTER SET utf8mb4 ;
```

-- 사용할 데이터베이스 선택

```
USE studydb;
```

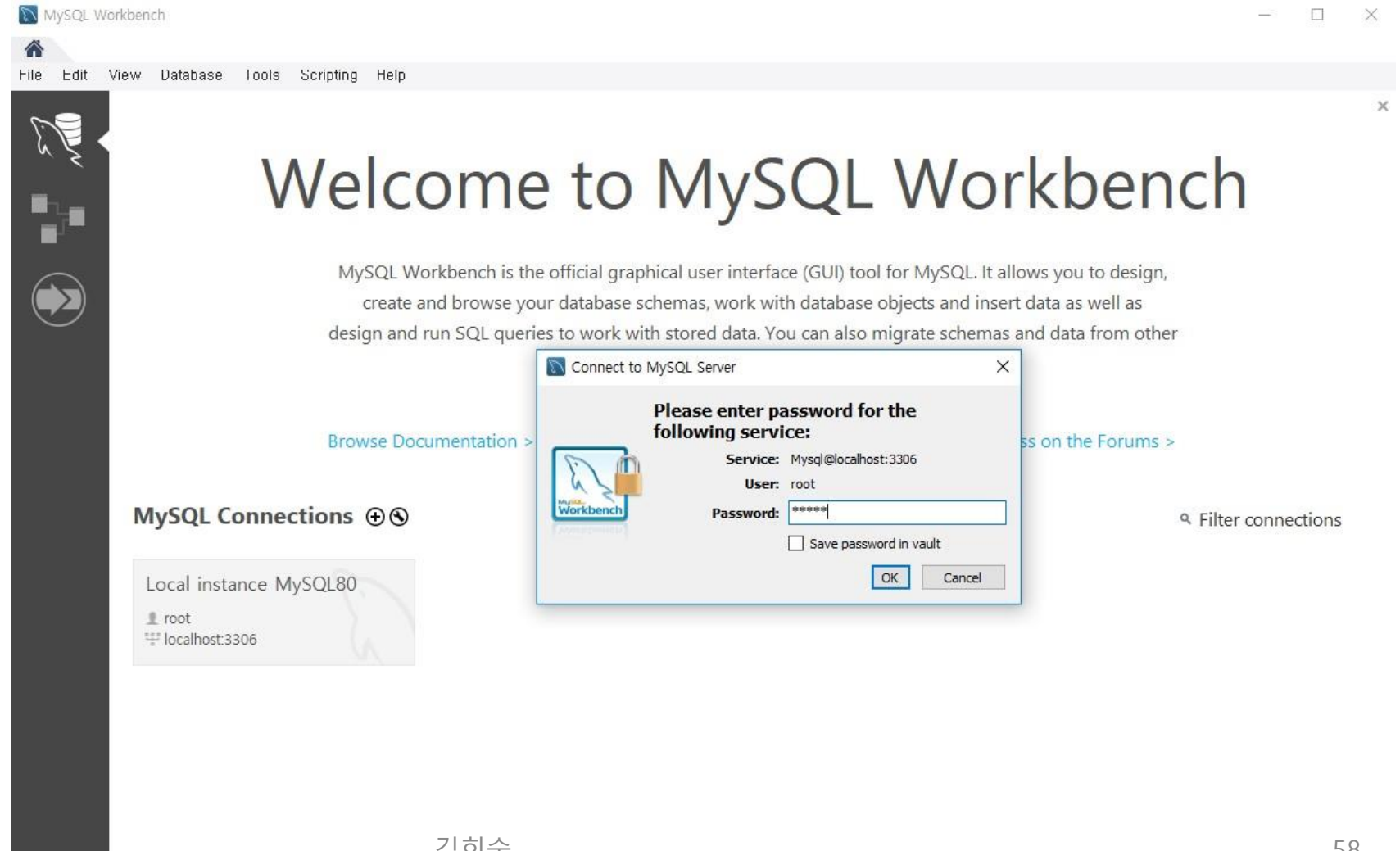
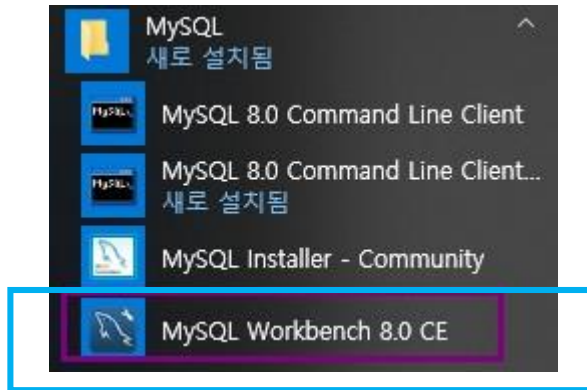
-- 데이터베이스 삭제

-- (MySQL)

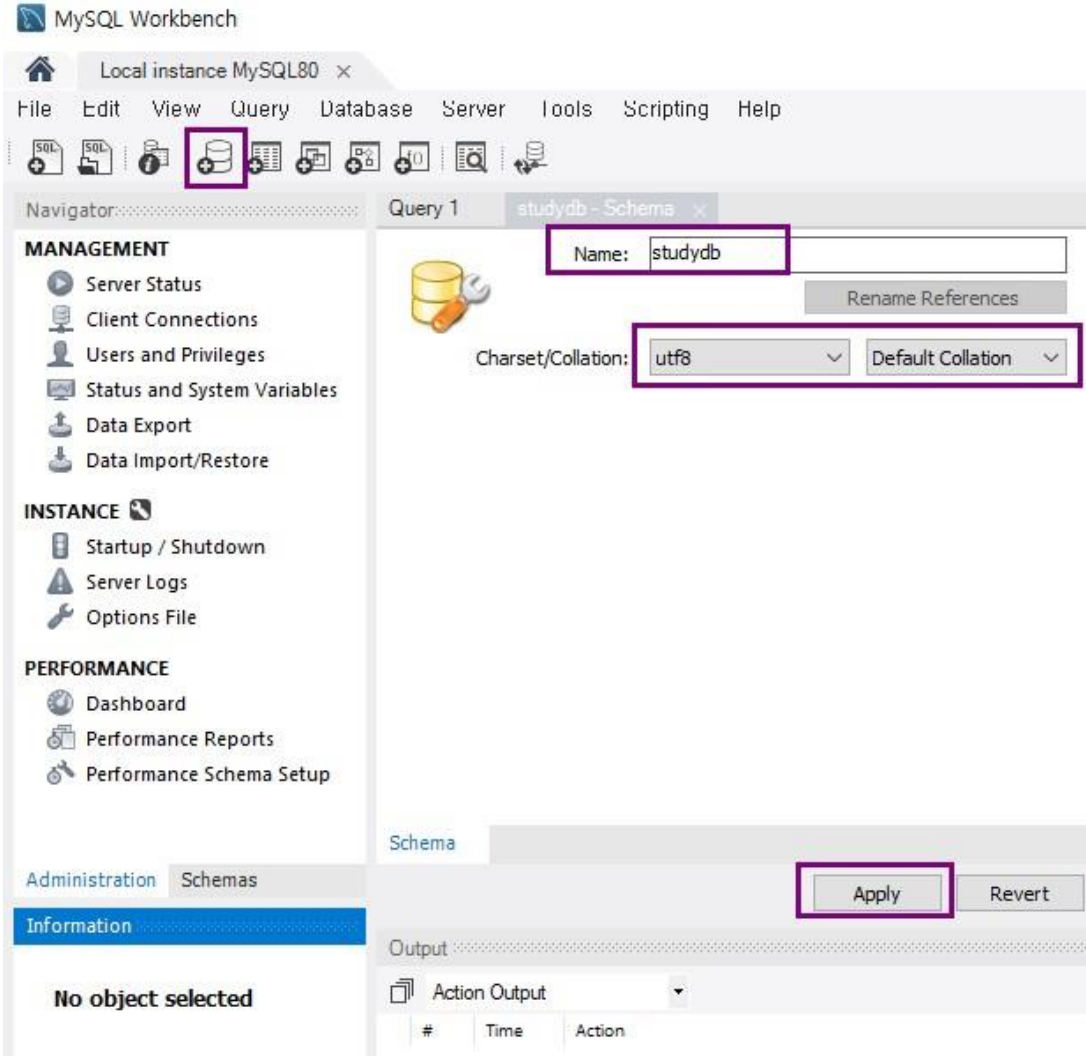
```
drop database testdb;
```

MySQL 실행

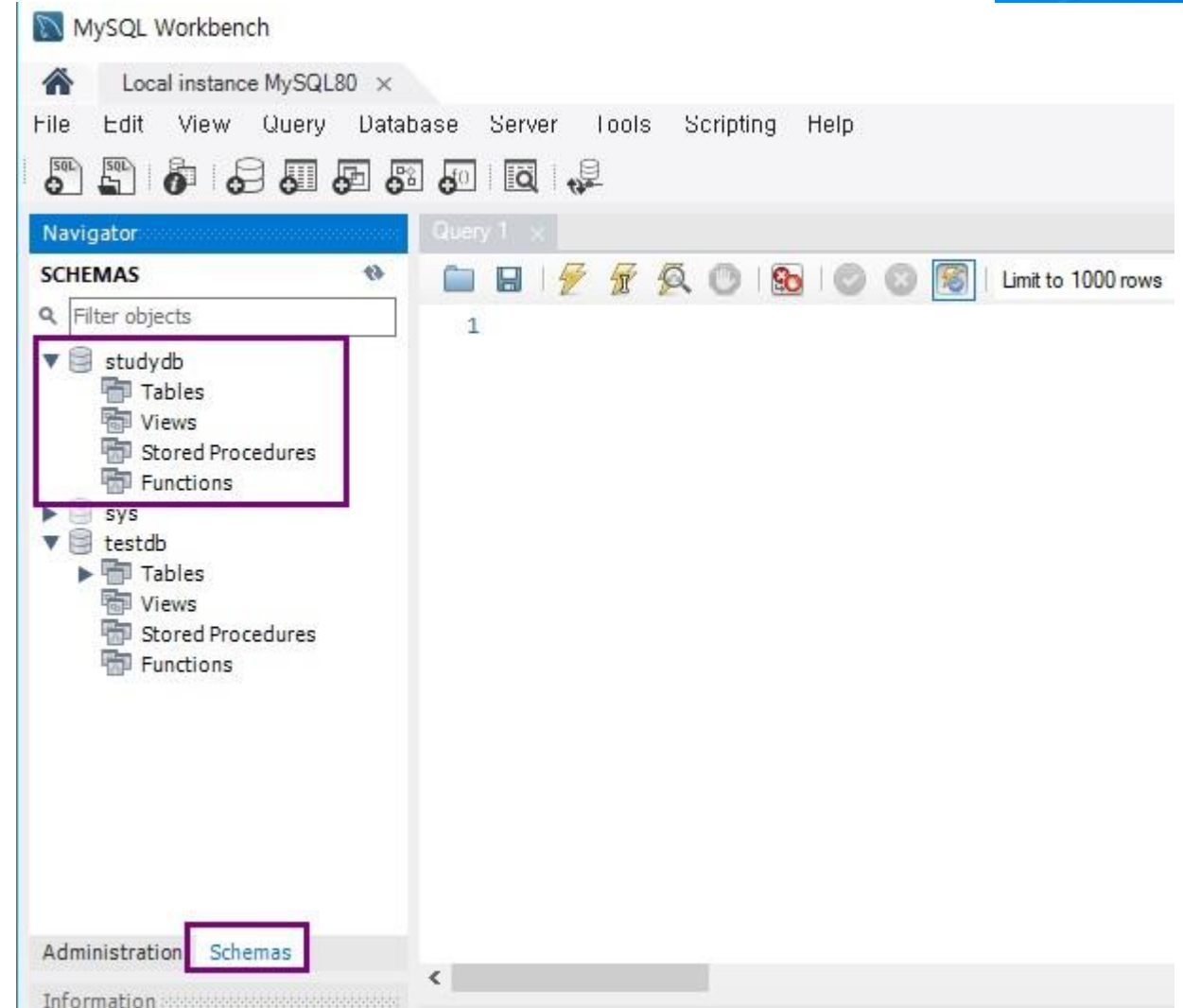
MySQL



MySQL 실습



MySQL



MySQL 실습

테이블 이름	열 이름	데이터 형식	NULL 유무	기본키
customer	cno	char(4)	NOT NULL	PK
	cname	varchar(10)	NOT NULL	
	city	varchar(20)		
	point	int		

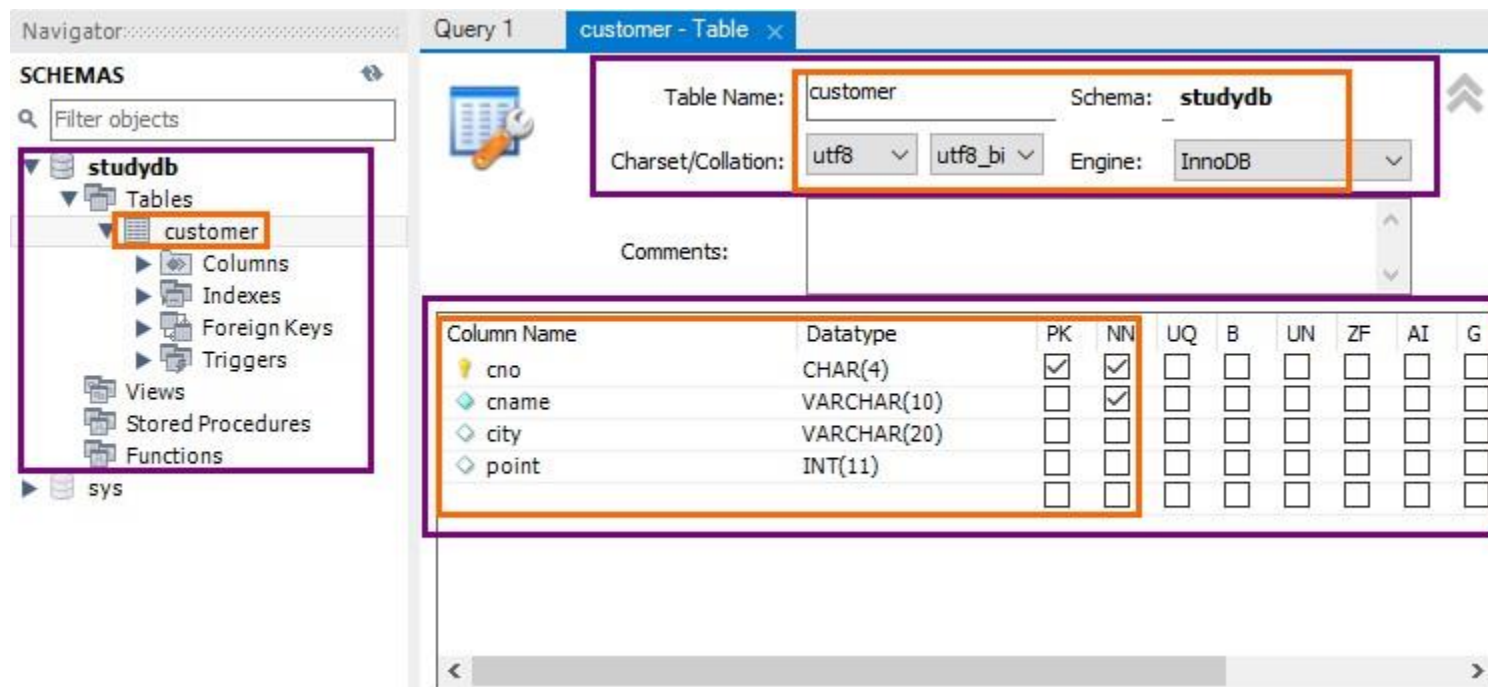


Table Name: customer Schema: studydb

Charset/Collation: utf8 utf8_bi Engine: InnoDB

Comments:

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G
cno	CHAR(4)	☒	☒	☐	☐	☐	☐	☐	☐
cname	VARCHAR(10)	☐	☒	☐	☐	☐	☐	☐	☐
city	VARCHAR(20)	☐	☐	☐	☐	☐	☐	☐	☐
point	INT(11)	☐	☐	☐	☐	☐	☐	☐	☐

MySQL 실습

테이블 이름	열 이름	데이터 형식	NULL 유무	기본키
customer	cno	char(4)	NOT NULL	PK
	cname	varchar(10)	NOT NULL	
	city	varchar(20)		
	point	int		

```
-- customer 테이블 생성
CREATE TABLE customer (
  cno      char(4)      NOT NULL   PRIMARY KEY ,
  cname    varchar(10)  NOT NULL ,
  city     varchar(20) ,
  point    int
);
```

MySQL 실습

테이블 이름	열 이름	데이터 형식	NULL 유무	기본키
customer	cno	char(4)	NOT NULL	PK
	cname	varchar(10)	NOT NULL	
	city	varchar(20)		
	point	int		

Result Grid   Filter Rows:				
	cno	cname	city	point
▶	c101	홍길동	서울	500
	c102	임꺽정	인천	300
	c103	박찬호	안양	800
	c204	신동엽	과천	350
	c205	정진우	고양	400

-- 데이터 입력

-- customer(cno, cname, city, point)

```
insert into customer values('c101','홍길동','서울',500);
insert into customer values('c102','임꺽정','인천',300);
insert into customer values('c103','박찬호','안양',800);
insert into customer values('c204','신동엽','과천',350);
insert into customer values('c205','정진우','고양',400);
```

-- 데이터 조회

```
select * from customer;
```

MySQL 실습

테이블 이름	열 이름	데이터 형식	NULL 유무	기본키
customer	cno	char(4)	NOT NULL	PK
	cname	varchar(10)	NOT NULL	
	city	varchar(20)		
	point	int		

Result Grid	cno	cname	city	point
▶	c101	홍길동	서울	500
	c102	임꺽정	인천	300
	c103	박찬호	안양	800
	c204	신동엽	과천	350
	c205	정진우	고양	400

* MySQL Workbench 사용

- 1) 한 줄 SQL 실행: <Ctrl + Enter>
- 2) 여러 줄 SQL 실행: <Ctrl + shift + Enter>



MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: Schemas

Filter objects

studydb

- Tables
- Views
- Stored Procedures
- Functions

sys

- testdb

Tables

- Views
- Stored Procedures
- Functions

Administration Schemas Information

Schema: studydb

Query 1

Limit to 1000 rows

```

1  /* customer 테이블 생성 */
2  create table customer (
3      cno char(4) NOT NULL PRIMARY KEY ,
4      cname varchar(8) NOT NULL ,
5      city varchar(20) ,
6      point int
7  );
8
9  /* customer(cno, cname, city, point) */
10 insert into customer values('c101', '홍길동', '서울', 500);
11 insert into customer values('c102', '임꺽정', '인천', 300);
12 insert into customer values('c103', '박찬호', '안양', 800);
13 insert into customer values('c204', '신동엽', '과천', 350);
14 insert into customer values('c205', '정진우', '고양', 400);
15
16 /* 데이터 조회 */
17 select * from customer;
    
```

Output

[실습] customer

테이블 이름	열 이름	데이터 형식	NULL 유무	기본키
customer	cno	char(4)	NOT NULL	PK
	cname	varchar(10)	NOT NULL	
	city	varchar(20)		
	point	int		

Result Grid					Filter Rows:
	cno	cname	city	point	
▶	c101	홍길동	서울	500	
	c102	임꺽정	인천	300	
	c103	박찬호	안양	800	
	c204	신동엽	과천	350	
	c205	정진우	고양	400	

-- MySQL 실습

/*

1단계: 스키마/데이터베이스 생성

studydb

2단계: 테이블 생성

customer

3단계: 데이터 입력

*/

-- MySQL

-- 1단계: 스키마 생성

CREATE SCHEMA studydb DEFAULT CHARACTER SET utf8 ;

-- 사용할 데이터베이스 선택

USE studydb;

-- 2단계: 테이블 생성

-- 테이블 삭제

drop table customer;

-- customer 테이블 생성

CREATE TABLE customer (
 cno char(4) NOT NULL PRIMARY KEY ,
 cname varchar(10) NOT NULL ,
 city varchar(20) ,
 point int
);

-- 데이터 입력

-- customer(cno, cname, city, point)

insert into customer values('c101','홍길동','서울',500);

insert into customer values('c102','임꺽정','인천',300);

insert into customer values('c103','박찬호','안양',800);

insert into customer values('c204','신동엽','과천',350);

insert into customer values('c205','정진우','고양',400);

-- 데이터 조회

select * from customer;